

ADVERSARIAL ATTACKS LEVERAGE INTERFERENCE BETWEEN FEATURES IN SUPERPOSITION

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ABSTRACT

The origins of adversarial examples (AExs) in neural networks (NNs) remain contested, with prevailing accounts attributing them either to irregularities in the decision landscape or to sensitivity to non-robust input features. We advance and substantiate a distinct thesis: adversarial vulnerability is a direct consequence of the *efficient* information encoding that NNs employ. Concretely, we establish that superposition—the encoding of more features than there are dimensions—produces arrangements of latent representations whose residual interference adversaries systematically exploit. Because each perturbation is fixed by this feature geometry, attack patterns become predictable from the representation alone. This single mechanism furnishes a unified account of two well-documented phenomena that prior explanations treat as separate: the transferability of adversarial attacks between models trained under similar regimes, and class-specific vulnerability. In synthetic settings with fully controlled superposition, we show that superposition *suffices* to induce adversarial vulnerability, formalise the mechanism through exact, closed-form expressions for the optimal perturbation, and confirm that the same behaviour persists in a vision transformer (ViT) trained on CIFAR-10 and in a ResNet-18; in a complementary orthogonal-representation setting we further establish that superposition is sufficient but *not* necessary, isolating algorithmic brittleness as a second, distinct failure mode and demonstrating that the resulting representation-aware analysis is actionable enough to construct attacks without gradients. Together, these results reframe adversarial vulnerability as an intrinsic byproduct of the representational compression through which networks encode information efficiently—not a symptom of flawed learning or of non-robust inputs—and establish the analysis of learned representations as a principled foundation for anticipating and countering attacks. We release our full implementation in an accompanying software repository.

1 INTRODUCTION

Despite a decade of intensive research on AExs (Szegedy et al., 2014; Goodfellow et al., 2014), no consensus has emerged on their fundamental cause. The field still lacks a mechanistic account of how minimal perturbations so drastically alter model predictions, and the absence of one carries concrete costs: without it, we can neither predict which perturbations will succeed, nor explain why attacks transfer between independently trained models, nor ground defences in principled reasoning rather than empirical heuristics. This paper supplies precisely such an account.

Existing explanations fall into two broad camps (Nakkiran, 2019). The *bug* perspective attributes vulnerability to flaws in the learning process (Schmidt et al., 2018; Nakkiran, 2019) or to the geometry of high-dimensional decision boundaries (Fawzi et al., 2016); work in this tradition supplies statistical bounds on quantities such as model capacity (Bubeck & Sellke, 2021) or Lipschitz constants (Hein & Andriushchenko, 2017), yet stops short of identifying which specific perturbations succeed. The *feature* perspective instead holds that AExs exploit predictive but non-robust statistical patterns in the data (Ilyas et al., 2019), recasting the debate around how the data relates to human perception; crucially, it treats these non-robust features as fixed properties of the data rather than as artefacts of the network’s encoding strategy. Neither account confronts how representational constraints interact with data semantics—and it is exactly this interaction, we demonstrate, that is decisive.

We resolve this divide through a mechanistic interpretability lens, establishing that adversarial vulnerability emerges from the interaction between architectural constraints and data semantics (*i.e.* the

human-interpretable properties of the data). We show that AExs arise from interference between the learned representations of NNs: adversarial attacks systematically exploit superposed latent structure—the very mechanism that ordinarily furnishes additional representational capacity—to construct perturbations that manipulate model outputs. This analysis exposes a coherent, end-to-end mechanistic pathway: input correlations constrain the feature arrangements, these arrangements determine the interference patterns, and the interference patterns in turn dictate both attack characteristics and transferability. The consequence is that the phenomenon becomes predictable—we explain which perturbations succeed and why they transfer across independently trained models—so that behaviour prior accounts treat as mysterious is recast as a direct consequence of the network’s representational geometry.

Our account builds on two recent theories of how networks encode and process information: the linear representation hypothesis (LRH) (Park et al., 2024) and the theory of superposition (Elhage et al., 2022). The LRH holds that *input features*—fundamental abstractions of the data—are encoded as linear directions in a network’s representation space. Superposition further posits that NNs represent substantially more such features than they have neurons, packing multiple features into shared dimensions at the cost of interference, so that perturbing one feature necessarily perturbs others in non-obvious ways. We establish that this interference is exactly the vulnerability that AExs exploit, and we trace what this reveals about adversarial phenomena more broadly. Our primary contributions are:

- **A unifying mechanistic framework.** We reconcile architectural constraints and feature properties as *joint* determinants of adversarial vulnerability. In controlled models we establish that data properties induce specific superposition geometries which in turn determine the exact adversarial perturbations a network admits, and we prove, through closed-form expressions for the optimal perturbation, that the resulting interference patterns account for both attack transferability and class-specific susceptibility.
- **Validation in realistic models.** We show that these findings persist in a ViT trained on CIFAR-10 with an engineered bottleneck layer that induces superposition—and reproduce them on a ResNet-18—confirming that the mechanism operates well beyond the synthetic setting and across architectures.
- **Sufficiency without necessity.** We establish that superposition is sufficient but *not* necessary for adversarial vulnerability, and in doing so isolate algorithmic brittleness as a second, distinct vulnerability mechanism that persists even under orthogonal representations.
- **Actionable, representation-aware attacks.** We demonstrate that a mechanistic understanding of learned representations enables the informed, gradient-free construction of adversarial attacks that nearly match gradient-based projected gradient descent (PGD), substantiating our argument that defences should be semantically informed rather than reliant on norm bounds alone.

2 BACKGROUND

We now formalise the two theoretical instruments that underpin our analysis: the LRH and superposition. Formally, let $\mathbf{x} \in \mathcal{X}$ denote the input to a NN and $\mathbf{h}^{(l)} \in \mathbb{R}^{d_l}$ the activation vector of the l -th layer.

Linear representation hypothesis (LRH). The LRH holds that NNs encode the variables of their computation—among them the semantic properties of their inputs—as linear directions in activation space, which then serve as the abstractions over which the network reasons (Park et al., 2024; Guerner et al., 2023). Two complementary considerations motivate this bias towards linear encodings: linear separability allows a network to recognise and manipulate features readily, and the dot products taken by subsequent-layer weights process such directional features efficiently. A substantial and growing body of evidence corroborates the hypothesis (Gurnee et al., 2023; Park et al., 2024). Let $\mathcal{C} = \{c_1, \dots, c_M\}$ denote a set of M semantically meaningful latent features (*e.g.* concepts such as “presence of shape” or “indoor vs. outdoor”). Formally:

Definition 1 (Linear Representation Hypothesis (LRH)) *A neural network layer with activations $\mathbf{h}^{(l)} \in \mathbb{R}^{d_l}$ satisfies the LRH if it represents the latent features $\mathcal{C} = \{c_1, \dots, c_M\}$ as linear directions*

$\{\mathbf{v}_j\}_{j=1}^M \subset \mathbb{R}^{d_l}$ such that:

$$\mathbf{h}^{(l)}(\mathbf{x}) \approx \sum_{j=1}^M a_j(\mathbf{x}) \mathbf{v}_j$$

where $a_j(\mathbf{x}) \in \mathbb{R}$ represents the activation magnitude of feature c_j , and $\mathbf{v}_j$ is the corresponding linear direction. The features are **linearly accessible** (Costa et al., 2025): inputs $\mathbf{x}_0, \mathbf{x}_1 \in \mathcal{X}$ that differ mainly in the value of feature c_j while holding other features approximately constant (i.e., $a_j(\mathbf{x}_1) - a_j(\mathbf{x}_0) = k$ and $|a_i(\mathbf{x}_1) - a_i(\mathbf{x}_0)| < \lambda$ for all $i \neq j$ and some small $\lambda > 0$) satisfy:

$$\mathbf{h}^{(l)}(\mathbf{x}_1) - \mathbf{h}^{(l)}(\mathbf{x}_0) \approx k \mathbf{v}_j$$

where k reflects the change in c_j .

Superposition and sparsity. NNs routinely encode far more latent features than there are dimensions available in activation space: $M > d_l$. Large language models, for instance, reference many more place names than they have residual-stream dimensions. Superposition is the principal framework for analysing this regime: the network employs an overcomplete, non-orthogonal set of feature directions $\{\mathbf{v}_j\}_{j=1}^M$, exploiting two facts—that $2^{\Theta(d\epsilon^2)}$ almost-orthogonal vectors (of pairwise cosine similarity below ϵ) fit within a d -dimensional space (Tkocz, 2012), and that sparse vectors can be recovered accurately after projection into lower-dimensional spaces (Elhage et al., 2022; Bereska & Gavves, 2024; Sawmya et al., 2025). This representational efficiency carries two consequences. First, *polysemanticity* (Scherlis et al., 2022; Lecomte et al., 2023) arises: an individual neuron contributes to several distinct features, so that its activation no longer corresponds to a single interpretable concept. Second, and central to our thesis, non-orthogonal feature directions necessarily induce *interference* between features—activating one feature partially activates others. A network can attenuate, but never eliminate, these effects through non-linear operations (e.g. ReLU, softmax) that disambiguate superposed features (Gurnee et al., 2023). It is precisely this residual, ineliminable interference that, as we show, adversarial attacks come to exploit.

Definition 2 (Superposition Hypothesis) A network layer represents features in superposition if:

1. The number of latent features exceeds layer dimensionality: $M > d_l$
2. There exists non-orthogonal feature directions: $\exists i, j$ with $i \neq j$ such that $\mathbf{v}_i \cdot \mathbf{v}_j \neq 0$
3. Latent feature activations are sparse: $\mathbb{E}_{\mathbf{x}}[\|\mathbf{a}(\mathbf{x})\|_0] \ll M$, where $\mathbf{a}(\mathbf{x}) = [a_1(\mathbf{x}), \dots, a_M(\mathbf{x})]^T$

The network trades representational capacity against feature interference by packing more features than dimensions.

Adversarial attacks. We now formalise adversarial perturbations and attacks, following (Szegedy et al., 2014; Costa et al., 2024):

Definition 3 (Adversarial attack) Let $\mathbb{B}_p(\mathbf{x}, \epsilon)$ denote the ϵ -ball around $\mathbf{x} \in \mathcal{X}$ in the ℓ_p -sense, e.g. $\mathbb{B}_\infty$ is the ℓ_∞ -ball. Any small δ that p -perturbs $\mathbf{x}$ to create an **adversarial input** $\mathbf{x}' \in \mathbb{B}_p(\mathbf{x}, \epsilon)$ such that $f(\mathbf{x}') \neq f(\mathbf{x})$ (untargeted) or $f(\mathbf{x}') = y_{\text{target}}$ (targeted) is known to be an **adversarial attack**. ℓ_∞ - or ℓ_2 -norm attacks are created via the PGD (detailed in App. A).

Definition 4 (Adversarial robustness) A classifier $g_w(\cdot)$ with parameters w is said to be **robust** at $\mathbf{x}$ with radius $\epsilon > 0$, if $g_w(\mathbf{x}') = g_w(\mathbf{x}) \forall \mathbf{x}' \in \mathbb{B}(\mathbf{x}, \epsilon)$.

3 SUPERPOSITION GEOMETRY DETERMINES ADVERSARIAL ATTACKS

We first establish whether, and precisely how, adversarial attacks exploit the interference rooted in the superposition of latent features. To render the mechanism both visible and testable, we work in a synthetic setting with fully controlled superposition—a setting whose identifiability, unavailable in naturalistic tasks, is what permits the exact, falsifiable predictions that follow. We organise the investigation around three questions:

1. Do adversarial perturbations exploit the interference between superposed features?
2. Do correlations in the input shape the geometric arrangement of superposed latent features?
3. Can shared latent geometry explain why attacks transfer between independently trained models?

Experimental setup. To address these questions under full experimental control, we design a synthetic task that instantiates the established LRH and superposition phenomena (Park et al., 2024; Elhage et al., 2022). The design satisfies four requirements by construction: (1) it furnishes an intuitive classification setting for studying AExs; (2) it encodes class concepts as linear directions, in accordance with the LRH; (3) it induces controlled superposition among these latent features; and (4) it retains *a priori* knowledge of the correspondence between inputs and superposed features. This last property—an identifiability that naturalistic tasks do not afford—is what makes the adversarial mechanism directly testable rather than merely conjecturable.

Concretely, we partition the input $\mathbf{x} \in \mathbb{R}^d$ into k equally sized groups $\mathbf{x} = [\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(k)}]$, where $\mathbf{x}^{(j)}$ comprises the p input features associated with class j . The task is to identify the group with the largest sum, $y = \operatorname{argmax}_{j \in \{1, \dots, k\}} \sum_{i=1}^p x_i^{(j)}$; each $x_i^{(j)}$ thus constitutes evidence for class j , and the p features encode distinct attributes of that class. Following Elhage et al. (2022), we sample $x_i^{(j)} \sim \text{Uniform}(0, 1)$ with sparsity S (the probability of setting an entry to zero).

A two-layer network compresses the input into m dimensions through a bottleneck encoder, $\mathbf{h} = \text{ReLU}(\mathbf{W}_e \mathbf{x} + \mathbf{b}_e) \in \mathbb{R}^m$ (with $m < k < d$), and reconstructs the class scores through a decoder, $\mathbf{z} = \mathbf{W}_d \mathbf{h} + \mathbf{b}_d \in \mathbb{R}^k$. Our primary configuration adopts cross entropy (CE) loss without ReLUs or biases—a deliberate choice made for analytic tractability, since the softmax bounds interference by exponentially suppressing contributions from interfering latent features and thereby isolates the cleanest linear setting in which the mechanism can be derived exactly. To confirm that our conclusions are not artefacts of this configuration, we replicate them under a second setup employing mean squared error (MSE) loss with ReLUs and biases. The representational challenge is thus to compress information from $d = k \times p$ dimensions into $m < k$ dimensions and then recover enough of it to identify the class with the largest sum—a compression that forces the k class directions into superposition.

Whereas naturalistic tasks involve features that encode nuanced concepts requiring discovery and interpretation, this setup supplies ground-truth labels for the latent features. We can therefore compute the true class after any perturbation by recomputing group sums, which lets us define an AEx rigorously as any perturbation satisfying two conditions:

1. The model’s prediction under perturbation changes (*e.g.* from class A to B)
2. The true class remains unchanged (*e.g.* still class A)

This criterion cleanly separates genuine AExs from perturbations that legitimately alter the ground truth (*i.e.* change which group has the highest sum)—a distinction that is simply unavailable in settings without ground-truth features. Attacks are generated using ℓ_∞ - and ℓ_2 -norm PGD.

Evaluation metrics. We interpret the columns of the encoder weight matrix $\mathbf{W}_e$ as the latent representations $\mathbf{v}_i$ of the corresponding input features x_i . Because input feature $x_i^{(j)}$ contributes exclusively to class j , the directions $\mathbf{v}_i$ (for $i \in \text{class } j$) align. This exact, known correspondence between input features and class-aligned latent directions is what allows us to trace, without ambiguity, how perturbations to the inputs of a given class, $\mathbf{x}^{(j)}$, propagate into latent space. For each successful PGD-generated AEx $\mathbf{x}_{adv} = \mathbf{x}_{orig} + \delta$, we measure the following quantities:

1. **Input perturbation profile (IPP):** The sign and magnitude of each δ_i , describing how different input features are perturbed in an attack.
2. **Latent attack alignment:** The similarity $\Delta \mathbf{h} \cdot \mathbf{v}_c$ between the *latent attack vector* $\Delta \mathbf{h} = \mathbf{h}_{adv} - \mathbf{h}_{orig}$ and class c ’s representation.
3. **Attack transferability:** Success rate of attacks generated on one model when applied to another.
4. **Robust accuracy:** The fraction of examples that remain correctly classified after perturbation δ with $\|\delta\|_p \leq \epsilon$ to a sample of inputs.

3.1 RESULTS

Do attacks exploit interference between latent features in superposition? Changing a sample’s class requires moving it across a decision boundary—but what determines the input perturbation that achieves this? Two competing intuitions frame the question. The *feature intuition* predicts that an attack simply amplifies the input features of the target class, whereas the *bug intuition* holds that adversarial perturbations modify inputs unpredictably, so that knowledge of the feature representations

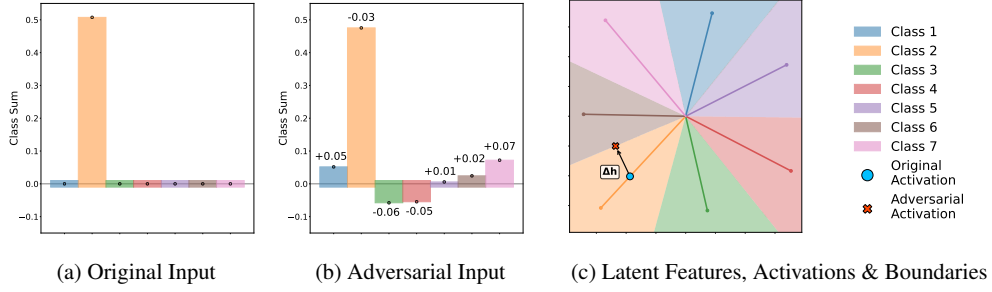


Figure 1: **An adversarial attack exploiting superposition geometry** ($k = 7, m = 2$). (a) The original sample. (b) The adversarially perturbed sample: its ground-truth class is unchanged, yet the model misclassifies it, and the sign and magnitude of every input perturbation are dictated by the configuration of latent representations rather than chosen arbitrarily. (c) The original and adversarial samples in activation space, where the arrows are the column vectors of $\mathbf{W}_e$ —the latent representations of the input features. Read together, the three panels establish that the perturbation is not arbitrary but is fully determined by the latent geometry.

confers no insight. Our results refute both and vindicate a third account: the perturbations are mediated by the latent geometry.

Fig. 1 depicts a representative AEx: an input of Class 2 (orange) and its perturbed counterpart (with IPP), which the model misclassifies as Class 6 (brown). The perturbation *appears* arbitrary—it plainly does not simply increase the target-class features, which already contradicts the feature intuition—yet it is far from random. It follows a precise pattern in which δ stands in systematic correspondence with the configuration of latent features, pointing decisively to the third account: *attacks are mediated by latent geometry*.

This structure is quantitatively precise: the IPP correlates strongly with the latent attack alignment. The sign and magnitude of the perturbation $\delta^{(j)}$ track how closely that class’s latent representation $\mathbf{v}^{(j)}$ aligns with the latent attack vector $\Delta\mathbf{h}$. Positively aligned representations ($\mathbf{v}^{(j)} \cdot \Delta\mathbf{h} > 0$) have their input features proportionally amplified, whereas negatively aligned representations are attenuated—precisely the behaviour Fig. 1(b) displays, in which the input perturbations are fixed by the dot products of Fig. 1(c) so as to drive the sample across the decision boundary.

To adjudicate directly whether PGD attacks exploit superposition geometry, we compare them against the theoretically optimal perturbations that, as we derive in Sec. 3.2, provably leverage superposition interference. For each configuration (k, m) we train 5 models under distinct random seeds, generate 1000 PGD attacks per model, and compute the cosine similarity between each successful perturbation and the corresponding optimum, retaining only perturbations of ℓ_2 norm at least ϵ to ensure they are meaningful; a norm-matched random baseline is measured against the same optimum. As Tab. 1 reports, PGD and optimal attacks are near-perfectly aligned across every dimensionality tested (cosine similarity 0.92–0.97), while the random baseline remains statistically indistinguishable from zero, with one-sample t -tests against that baseline yielding $p < 10^{-10}$ in every configuration. Since the optimal attacks provably exploit interference and the PGD attacks match them almost exactly, we conclude that PGD attacks likewise exploit superposition interference.

Finding: Adversarial attacks systematically exploit interference between superposed features. Successful PGD attacks are predictable from the specific superposition geometry, rather than random.

Table 1: Alignment between PGD-discovered attacks and the theoretically optimal (interference-exploiting) perturbations across superposition configurations (k, m) . The cosine similarity of PGD attacks to the theoretical optimum is near-perfect (0.92–0.97) and vastly exceeds a norm-matched random baseline—which is statistically indistinguishable from zero—at every dimensionality tested. Values are mean $\pm$ std over 1000 attacks per condition; every p -value falls below 10^{-10} .

k	m	Cosine Sim. (PGD vs Theory)	Cosine Sim. (Random Baseline)
6	2	0.97 ± 0.02	0.00 ± 0.02
30	10	0.96 ± 0.00	0.00 ± 0.01
90	30	0.92 ± 0.00	0.00 ± 0.00

Do input correlations determine latent feature geometry? Having shown that attacks are dictated by the feature geometry, we now establish what determines that geometry in the first place. By systematically manipulating the correlation structure of the training data, we probe how input correlations constrain the degrees of freedom in the learned representations, introducing three correlation patterns of increasing constraint strength. Under the *uncorrelated* condition, features are sampled independently, as in the previous subsection. Under *paired* correlations, each feature in class i is coupled to a feature in class j so that the two co-activate. Under *global* correlations, each sample carries a random phase that induces a cyclic activation pattern: classes adjacent in index share similar activation probabilities and therefore co-activate frequently, whereas classes far apart in index rarely activate together.

Fig. 2 shows how these correlations govern the arrangement of the $\mathbf{v}^{(j)}$. We quantify geometric similarity by forming, for each model, the matrix of pairwise cosine similarities over all feature pairs and then correlating these matrices across random seeds, so that a higher correlation signals a more similar geometric arrangement. Testing 25 seed pairs per condition, we obtain, for the $m = 2, k = 7$ setup, geometric similarities of 0.23 ± 0.14 (uncorrelated), 0.47 ± 0.32 (paired), and 0.88 ± 0.06 (global) (mean $\pm$ std). The relationship is strictly monotonic in correlation strength: uncorrelated data yields highly variable geometries across seeds, paired correlations partially constrain the arrangement, and global correlations force near-identical geometries. Pairwise two-sample t -tests across all three modes, Bonferroni-corrected ($\alpha = 0.05$ over 3 comparisons), confirm significance at $p < 10^{-5}$ for every comparison, and the same ordering holds across the broader range of (k, m) configurations reported in App. C.3—establishing that the effect is a general property of the correlation structure rather than an artefact of the specific configuration.

The governing mechanism is *interference avoidance*: features that co-activate frequently are arranged so as to minimise their mutual interference, and the strength of the correlation dictates how tightly this constrains the geometry—uncorrelated data admits any class-separating arrangement, paired correlations fix the relative positions of coupled features, and global correlations determine a unique geometry up to rotation.

Finding: Input correlations constrain feature geometry. Stronger correlations reduce the geometric degrees of freedom, forcing different initialisations to converge to similar arrangements.

Does shared geometry explain attack transferability? Having established that input correlations induce consistent latent geometries across initialisations, we now close the causal chain by testing whether the resulting shared interference patterns explain attack transferability. If an attack is tailored to a specific interference pattern, it should transfer successfully only between models whose geometric arrangements coincide—a sharp, falsifiable prediction that our data confirm.

Reusing the seed pairs from the preceding analysis, we generate attacks on source models and measure their success on target models within each correlation condition (25 transfer measurements per condition); all models exceed 95% clean accuracy. Transfer rates track geometric similarity closely. For the $k = 7, m = 2$ configuration (Fig. 2), globally correlated data yields $94\% \pm 12\%$ transfer (mean $\pm$ std), uncorrelated data only $18\% \pm 9\%$, and paired correlations an intermediate $32\% \pm 20\%$ —a monotonic ordering that mirrors the geometric-similarity results exactly, so that shared geometry and transferability rise and fall in lockstep.

The mechanism follows directly. Each perturbation component amplifies or suppresses a feature, producing constructive interference that pushes the representation across a decision boundary. Applied to a model with a different feature arrangement, the same perturbation causes the previously constructive contributions to cancel or interfere destructively, and the attack fails accordingly.

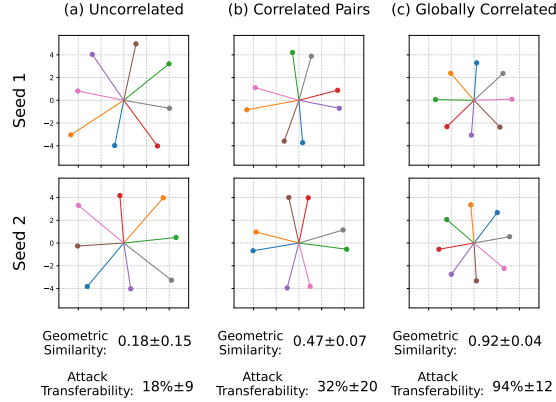


Figure 2: Stronger input correlations yield more consistent latent geometries across initialisations, driving attack transferability from 18% (uncorrelated) to 94% (global)—a more than fivefold increase. Error bars show standard deviations.

Finding: Attack transferability is governed by shared interference patterns.

Does reducing superposition suppress these attacks? If superposition creates vulnerability through interference, then removing superposition should abolish it. Three complementary interventions, each curtailing interference in a distinct way, jointly confirm this prediction. First, setting $m = k$ leads the network to represent every class direction $\mathbf{v}^{(j)}$ orthogonally; in this configuration any perturbation that flips the model’s prediction necessarily flips the ground-truth class, since moving a sample from class A to class B requires the sum of class B’s features to exceed class A’s—genuinely converting it into a class B sample. Consistent with this, we observe zero successful adversarial examples across 1000 attempts at every ϵ tested. Second, to disentangle superposition from capacity, we hold m fixed and vary k : robust accuracy decreases monotonically with the superposition pressure k/m (App. Tab. 5), establishing that vulnerability scales with the degree of interference rather than with capacity alone. Third, to verify that attacks concentrate specifically on superposed features, we constrain one class’s latent vector $\mathbf{v}^{(\perp)}$ to remain orthogonal to all others throughout training; when attacks are then generated between classes that remain in superposition (e.g. from class a to b with $\mathbf{v}_a \not\perp \mathbf{v}_b$), the inputs corresponding to the orthogonal class are left entirely unperturbed (Fig. 3). Taken together, these three interventions establish that superposition is not merely correlated with the vulnerability but causally responsible for it.

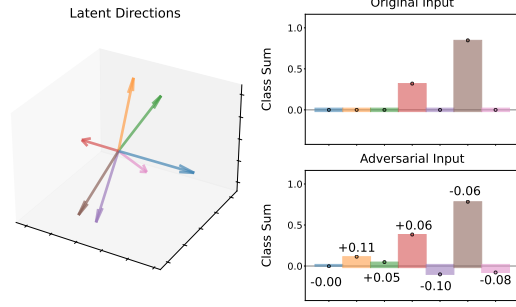


Figure 3: An adversarial attack (from class 5 to class 3) leaves the input features of a class held orthogonal to the others (class 1) entirely unperturbed, confirming that attacks act exclusively on features in superposition.

Finding: Adversarial attacks expend their budget precisely on the features held in superposition.

3.2 FORMAL ANALYSIS

We now formalise the mechanism underlying these empirical observations. In the linear setting the relationship between superposition interference and adversarial vulnerability admits an exact derivation, elevating our findings from observational correlation to provable structure and supplying the theoretical optimum against which the PGD attacks of Tab. 1 were compared. We adopt the notation of the experimental setup: $\mathbf{W}_e \in \mathbb{R}^{m \times d}$ induces superposition through its columns $\mathbf{v}_i$, the decoder coincides with the encoder transpose (as we observe empirically), and the logit for class j is $z_j = \mathbf{v}_j^\top \mathbf{h}$.

Proposition 1. *The optimal input perturbations δ that maximise movement from class j to class k under constraint $\|\delta\|_2 = \epsilon$ satisfy $\delta \propto \mathbf{W}_e^\top \mathbf{n}$, where $\mathbf{n} = (\mathbf{v}_k - \mathbf{v}_j)$ is the normal to the decision boundary between classes.*

Proof Sketch. To move from class j to k , we maximise the logit margin $z_k - z_j = \mathbf{n}^\top (\mathbf{h} + \Delta \mathbf{h})$, where $\mathbf{n} = (\mathbf{v}_k - \mathbf{v}_j)$ and $\Delta \mathbf{h} = \mathbf{W}_e \delta$. This becomes $\max_{\|\delta\|_2 = \epsilon} \delta^\top \mathbf{W}_e^\top \mathbf{n}$. By Cauchy-Schwarz, this is maximised when $\delta \propto \mathbf{W}_e^\top \mathbf{n}$. Full proof in App. B. $\square$

Corollary 1 (Interference Drives Vulnerability) *The adversarial perturbation magnitude for feature i is $|\delta_i| \propto |\mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j)|$, directly proportional to the differential interference between feature i and the class representations.*

This makes explicit how superposition creates adversarial vulnerability: each input feature i is perturbed in proportion to its interference with the class representations. Under superposition the feature directions are non-orthogonal, so that even semantically unrelated features interfere with the class decision, and adversarial perturbations exploit precisely these dependencies to manipulate the output. When the representations are orthogonal, the interference vanishes and the vulnerability vanishes with it—exactly the behaviour observed in the $m = k$ intervention above.

Proposition 2. *Models with feature representations related by orthogonal transformation $\mathbf{Q}$ (where $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$) share identical optimal attack directions in input space.*

Proof Sketch. Under transformation $\mathbf{v}'_i = \mathbf{Q}\mathbf{v}_i$, optimal perturbations remain invariant: $\delta'_i \propto (\mathbf{v}'_i)^\top (\mathbf{v}'_k - \mathbf{v}'_j) = \mathbf{v}_i^\top \mathbf{Q}^\top \mathbf{Q} (\mathbf{v}_k - \mathbf{v}_j) = \mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j) = \delta_i$. Full proof in App. B. $\square$

Together with our empirical findings, these propositions establish a complete **mechanistic pathway**: (i) input correlations constrain the feature arrangements in superposition, (ii) these geometric arrangements determine the interference patterns, (iii) the interference patterns dictate the optimal perturbations via $\delta \propto \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$ (Proposition 1), and (iv) shared geometric constraints yield similar interference patterns across models, which enables transferability (Proposition 2):

Correlations $\xrightarrow{\text{constrain}}$ **Feature Geometry** $\xrightarrow{\text{determines}}$ **Interference Patterns** $\xrightarrow{\text{enable}}$ **Transferability**

4 ATTACKS IN REALISTIC MODELS

Having isolated the mechanism in the synthetic setting, we now demonstrate that it operates in realistic models—a ViT (Dosovitskiy et al., 2020) trained on CIFAR-10 (Krizhevsky, 2009), together with a ResNet-18 (He et al., 2016) in App. D.2.1. Inserting a bottleneck layer immediately before the final classification output again forces the latent class features into superposition, which lets us test three predictions of the toy-model analysis in a naturalistic model: (i) that the degree of superposition governs attack transferability and robustness, (ii) that a consistent arrangement of the superposed representations emerges across initialisations, and (iii) that attacks exploit feature interference, exactly as in the toy models. All three predictions are borne out.

Setup. We train our ViT (6 layers, patch size 4, residual stream $d = 512$) on CIFAR-10 across five random seeds. After training, we replace the original classification head with a bottleneck: a linear encoder that compresses the pre-classification activations into an m -dimensional space, followed by a linear decoder that expands them back to the $k = 10$ CIFAR-10 classes. Trained with CE loss and no ReLU—deliberately mirroring the configuration of Sec. 3—this bottleneck forces the k class representations into m -dimensional superposition. The weights of the base ViT are frozen while the bottleneck is trained, and we sweep the bottleneck dimension $m \in \{2, 3, 5, 10\}$ to modulate the degree of superpositional compression. AExs are generated for each bottlenecked model with both ℓ_∞ - and ℓ_2 -norm PGD, using a small step size and many iterations and terminating as soon as an attack changes the model’s prediction (full details in App. D.1). To control for performance differences across bottleneck capacities, robust accuracy is normalised by each model’s clean accuracy, and attack transferability is measured exactly as defined in Sec. 3.

Results. The base models attain a mean clean accuracy of 81%, and the clean accuracy of the bottlenecked models declines predictably with compression—54%, 67%, 76%, and 81% for $m = \{2, 3, 5, 10\}$ —confirming that the bottleneck imposes the intended representational pressure.

Three observations reproduce the toy-model results and thereby establish that the mechanism carries over intact to this realistic setting. First, Fig. 4 (right) shows that as the bottleneck m decreases—that is, as superposition pressure rises—normalised robust accuracy falls while attack transferability across initialisations rises. Both trends follow directly from the mechanism established in Sec. 3. The decline in robust accuracy reflects the denser packing of class representations, which lets small input perturbations induce widespread interference across the compressed features. The rise in transferability follows from our earlier finding that greater superposition reduces the degrees of freedom in the feature geometry: a more tightly constrained representational space admits fewer viable arrangements for the class features, so that distinct initialisations converge to more similar geometries—and hence to more shared interference patterns—which in turn yields higher attack transferability. Results across the full range of perturbation magnitudes, reported in App. D.1.1, confirm that these trends hold at every ϵ .

Second, and reinforcing this, Fig. 4 (left) shows that the m -dimensional class representations within the bottleneck exhibit consistent ordering and clustering across models—a consistency that, exactly as in Sec. 3, indicates that inherent data correlations guide the formation of the superposed geometry. These groupings frequently recover genuine semantic similarities between classes (e.g. the ‘cat’ and ‘dog’ representations cluster together), demonstrating that the correlation-driven arrangement is semantically meaningful rather than incidental. Third, when AExs generated for a bottlenecked model are passed through the corresponding base ViT, they induce similar relative logit changes (i.e. $\text{logit}_{\text{base}}(\mathbf{x}_{\text{adv}}) - \text{logit}_{\text{base}}(\mathbf{x}_{\text{orig}})$), so that both models respond consistently to the same perturba-

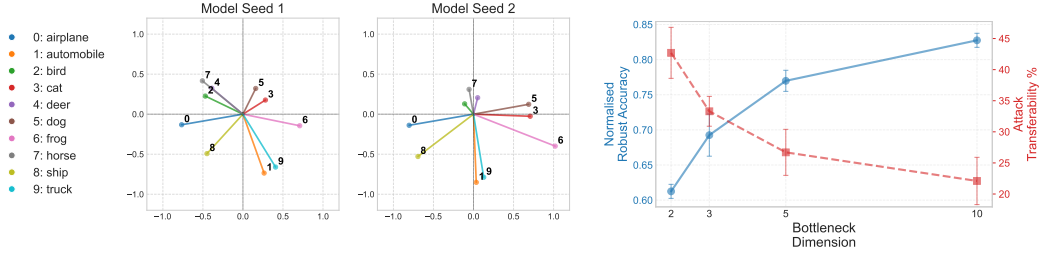


Figure 4: Left: the structure of the CIFAR-10 class representations remains consistent across models trained from different seeds, indicating a shared, correlation-driven geometry. Right: attack transferability and normalised robust accuracy as the bottleneck dimension increases—transferability falls and robustness rises monotonically as superposition is relaxed, precisely as the toy-model mechanism predicts.

tions. The perturbations are therefore not an artefact of the bottleneck but act on representations already present in the base model; absent the added compression, however, they do not on their own cross the decision boundary to produce misclassifications. Together, these three observations confirm that the superposition mechanism—not any idiosyncrasy of the synthetic task—drives adversarial vulnerability in a naturalistic model.

5 ADVERSARIAL VULNERABILITY VIA ALGORITHMIC BRITTLINESS

The preceding sections established that feature superposition *suffices* for adversarial vulnerability. We now establish that it is *not necessary*, by exhibiting a setting in which features are represented orthogonally—not in superposition—yet the model remains vulnerable to attack. This case simultaneously delimits the superposition mechanism and substantiates the paper’s broader thesis: that the analysis of learned feature representations is a general instrument for understanding robustness. We pursue three questions: *how do such attacks work mechanistically, can this mechanistic understanding be used to construct attacks directly, without gradients, and does naive robustness certification alter the underlying failure mode or merely raise the attack budget required to exploit it?*

Setup. We train a linear encoder followed by an MLP on one-hot integers (a, b) to predict $(a + b) \bmod P$. As detailed in App. E and Nanda et al. (2023), the model discovers a trigonometric algorithm for the task. Within this learned algorithm¹ the encoder projects inputs onto non-basis-aligned linear representations encoding $\sin(\omega_k a)$ and $\cos(\omega_k a)$ at the key frequencies $\omega_k = 2\pi k/P$, and the MLP then composes $\cos((a + b)\omega_k)$ through sum-angle identities to reach 100% accuracy. Crucially, these features are represented *orthogonally*—that is, not in superposition—which makes this task a clean and decisive test of necessity. We generate AExs by relaxing the one-hot constraint and perturbing only the input a with PGD, thereby driving the input off the training manifold.

Attacks exploit the learned key frequencies. Fourier analysis of the successful perturbations δ (App. Fig. 6) reveals that their dominant frequencies match the learned algorithm’s key frequencies ω_k precisely, and that exactly 0% of attacks transfer to models with different ω_k (e.g. those trained from different seeds)—the sharpest possible signature that the attack is keyed to the specific learned algorithm. The frequency-aligned perturbations induce misclassification by corrupting the sinusoidal values read off from the encoder features: the perturbation δ transforms the one-hot $\mathbf{x}_a$ into a dense vector $\mathbf{x}'_a$, and the contributions from the now non-zero inputs corrupt those sinusoidal values, thereby disrupting the MLP’s downstream computation. Because the algorithm relies on the constructive interference of every ω_k at the logits, disrupting a single ω_k is insufficient; a successful attack must instead find a δ that maximally alters multiple frequency components simultaneously. A detailed account of this mechanism is given in App. E.1.

Analytic construction of attacks. This mechanistic understanding enables the gradient-free, *informed* construction of attacks as composite sinusoidal waves at the matching frequencies ω_k . On $0 + 0 \bmod P$, such attacks succeed at $\|\delta\|_2 = 0.22$ —within roughly $1.3\times$ of the PGD budget (0.17)

¹We refer to the specific computation the network executes to solve the task as the *model’s learned algorithm*. Because this algorithm depends on extracting precise values from learned feature representations (frequencies), it remains *brittle* under small perturbations to the bases; we term this *algorithmic brittleness*.

and far below every naive alternative (single-index 0.97, ℓ_∞ noise 2.04, uniform 6.20). That the informed attack nearly matches gradient-based PGD while dramatically outperforming every uninformed baseline confirms that we have identified the correct mechanism, and establishes that a representation-aware view of vulnerability yields concrete, constructive attack recipes rather than merely post-hoc explanations.

Do defences mitigate this algorithmic vulnerability? We apply certified training against ℓ_∞ perturbations (Müller et al., 2023). At fixed attack budgets, certification improves robust accuracy markedly from the pre-robustified to the post-robustified model: 0% $\rightarrow$ 85% at $\epsilon_\infty = 5e-3$, 0% $\rightarrow$ 96% at $2e-3$, and 76% $\rightarrow$ 100% at $1e-3$. Yet under PGD the robustified model fails through the very same frequency-based brittleness, now merely at larger perturbation norms. Certification therefore raises the required attack budget without addressing the underlying failure mode—a diagnosis that directly motivates defences targeting the identified semantics (here, frequency manipulation) rather than relying on broad norm bounds alone.

Relation to the paper’s thesis. This section contributes two results that integrate directly with the paper’s broader argument. First, superposition is *sufficient but not necessary*: orthogonal representations can themselves remain vulnerable to attack, so the space of representation-driven vulnerabilities is strictly larger than superposition alone. Second, viewing AExs through the lens of *architecture–semantics interaction* is actionable, guiding both the design of attacks (via representation-informed perturbations) and the development of targeted defences.

These two vulnerability mechanisms are, moreover, linked by a shared dependence on sparsity assumptions: upstream violations of sparsity (e.g. noise or superposed features) can deliver effectively ‘pre-perturbed’ inputs to downstream orthogonal circuits, thereby activating the same frequency-based failure.

6 RELATED WORK

Superposition & latent representations. Research on superposition has characterised the mechanisms of feature interference: Nanda (2023) distinguish *representational* from *computational* superposition, and Gurnee et al. (2023) differentiate *alternating* from *simultaneous* interference. Data correlations demonstrably shape the resulting arrangements—pairs of correlated features become orthogonal (Elhage et al., 2022), drive the formation of superposition (Chan, 2024), and produce semantic clustering (Prieto et al., 2025a)—and it is exactly this correlation-to-geometry dependence that our framework renders mechanistically precise and, for the first time, links directly to adversarial vulnerability. Beyond superposition, representation geometry is further shaped by spectral bias (Rahaman et al., 2019), neural collapse (Kothapalli, 2023), and the choice of optimisation objective (Casper, 2023). Sparse autoencoders (SAEs) have been applied to vision models to disentangle features held in superposition (Lim et al., 2024; Joseph et al., 2025).

Adversarial vulnerability. The literature is organised around an attack–defence dichotomy. Attacks (Szegedy et al., 2014) employ imperceptible perturbations to maximise model error, including FGSM (Goodfellow et al., 2014) and PGD (Madry et al., 2018), while adversarial robustness denotes the maintenance of performance under such attacks. Defences range from adversarial training (Madry et al., 2018) to certified training (Wong & Kolter, 2018; Palma et al., 2024), the latter furnishing formal guarantees against perturbations within specified input bounds.

The role of input & latent features in adversarial robustness. Vulnerability has been attributed to predictive but brittle ‘non-robust’ input features (Ilyas et al., 2019; Goh, 2019) and to learning shortcuts (Li et al., 2023), with perturbation characteristics known to vary across datasets (Maiya et al., 2021). Elhage et al. (2022) conjecture a link between superposition and AExs, a possibility subsequently debated by Casper (2023); our results settle this debate, establishing the link mechanistically and demonstrating it in both synthetic and realistic models. A tension between representational efficiency and adversarial robustness has likewise been observed (Barsbey et al., 2025), and our account pinpoints interference in superposition as its concrete mechanistic source. Zhang et al. (2021) show that adversarial perturbations push latent representations across decision boundaries, and Ganeshan et al. (2019) find that PGD targets the final layers and accordingly propose feature-based attacks. Transferability is limited by representational discrepancies between models (Li et al., 2023; Wang et al., 2024), and decorrelating features reduces transfer (Wiedeman & Wang, 2022)—an

observation our framework explains directly, since it identifies shared latent geometry as the very substrate of transferability.

7 DISCUSSION & CONCLUDING REMARKS

We have established that adversarial attacks exploit the interference patterns arising from the geometry of superposed features in NNs. Across toy models, ViTs, and ResNets, our experiments demonstrate that data properties—specifically correlations and sparsity—induce distinct superposition geometries that in turn create predictable adversarial vulnerabilities, and that these geometric arrangements and their attendant interference render phenomena such as attack transferability and class-specific susceptibility predictable rather than mysterious. Our formal analysis makes the mechanism exact; our orthogonal-representation study shows that it is sufficient without being necessary; and our informed, gradient-free attacks show that it is actionable. Taken together, these results reframe adversarial vulnerability as an intrinsic consequence of the very mechanism by which networks efficiently encode vast numbers of features through superposition, and they establish the analysis of learned representations as a principled foundation for anticipating and countering attacks.

Limitations & future work. Our conclusions are grounded principally in controlled models—a scope we adopt deliberately, because it is precisely what renders the mechanism identifiable and its predictions testable. These analyses supply a mechanistic lens on attacks that we have shown transfers from synthetic tasks to ViTs and ResNets; the broader adversarial-robustness literature nonetheless indicates that the precise vulnerability mechanism will vary with attack and model type, and constrained attack classes (*e.g.* universal perturbations, jailbreaks, patch attacks) may be unable to activate the specific feature combinations required to exploit superposition interference.

Several concrete directions follow. Particularly promising avenues include quantifying the ‘adversarial cost’ of superposition, characterising its relation to accuracy–robustness trade-offs, and determining how robust training reshapes the superposition geometry. Because our controlled settings place class features in engineered superposition, whereas real networks involve interference among unknown, unlabelled features distributed across many layers, we intend to employ SAEs (Bricken et al., 2023) to extend the analysis to this regime at scale; the preliminary results in App. F already indicate that this is a viable and promising avenue.

ETHICS STATEMENT

Owing to its investigative nature and generality, our mechanistic interpretability framework is not anticipated to carry any specific negative impact or immediate societal consequence. Our experiments draw exclusively on publicly available datasets (CIFAR-10) and synthetic toy models, and involve no human subjects or personally identifiable information. The gradient-based attack methods we employ are already well established in the literature; our work introduces no new attack capability, contributing instead a mechanistic understanding of existing phenomena. We declare no conflicts of interest. By elucidating the relationship between superposition and adversarial vulnerability, we aim to inform future research that balances model capability against robustness, and thereby to contribute to more trustworthy AI systems suitable for safe deployment in critical applications.

REPRODUCIBILITY STATEMENT

We include source code for all experiments in the supplementary materials. The toy model experiments (Sec. 3) are documented in full, with complete architectural details, training procedures, evaluation metrics, and data-generation processes. For the ViT experiments (Sec. 4) we provide the architecture and training procedures, with further details in App. D.1. The modular arithmetic experiments (Sec. 5) specify the network architecture, data encoding, and learned trigonometric algorithm, with additional details in App. E.2. All experiments were conducted on an AMD Ryzen Threadripper 3970X 32-Core with 256GB RAM and an RTX 3090. The supplementary ResNet-18 and SAE experiments are documented in App. D.2.1 and App. F, respectively.

LLM USAGE STATEMENT

This work used LLMs to assist with prior literature search, to automate LaTeX table generation from CSV files (all numerical values were subsequently verified), and to suggest alternative phrasings that improve manuscript clarity. All substantive research contributions and conclusions remain the original work of the authors.

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APPENDIX

A GENERATION OF ADVERSARIAL EXAMPLES

We generate AExs using PGD, the standard iterative first-order attack (Madry et al., 2018). For untargated attacks it maximises the classifier’s loss $\mathcal{L}(f(\mathbf{x}'), y_{\text{true}})$; for targeted attacks it minimises $\mathcal{L}(f(\mathbf{x}'), y_{\text{target}})$; in either case it proceeds through the update rule:

$$\mathbf{x}'^{(k+1)} = \Pi_{\mathcal{S}} \left(\mathbf{x}'^{(k)} \pm \alpha \mathbf{g}_k \right) \quad (1)$$

(+ for maximisation, – for minimisation), where $\mathbf{x}'^{(0)}$ is the initial perturbed input, α the step size, and $\mathbf{g}_k$ the (normalised) gradient $\nabla_{\mathbf{x}} \mathcal{L}(f(\mathbf{x}'^{(k)}), y_{\text{class}})$ (with y_{class} equal to y_{true} or y_{target}); the operator $\Pi_{\mathcal{S}}$ projects onto the ϵ -ball $\mathcal{S} = \{\mathbf{x}' \mid \|\mathbf{x}' - \mathbf{x}\|_p \leq \epsilon\}$ and clips to the input domain. We consider the following ℓ_p norms:

1. ℓ_{∞} : Constraint $\|\delta\|_{\ell_{\infty}} = \max_i |\delta_i| \leq \epsilon$. $\mathbf{g}_k$ is $\text{sign}(\nabla_{\mathbf{x}} \mathcal{L}(\cdot))$. $\Pi_{\mathcal{S}}$ clips each $\mathbf{x}'_i$ to $[\mathbf{x}_i - \epsilon, \mathbf{x}_i + \epsilon]$.
2. ℓ_2 : Constraint $\|\delta\|_{\ell_2} = \sqrt{\sum_i \delta_i^2} \leq \epsilon$. $\mathbf{g}_k$ is $\nabla_{\mathbf{x}} \mathcal{L}(\cdot) / \|\nabla_{\mathbf{x}} \mathcal{L}(\cdot)\|_{\ell_2}$. $\Pi_{\mathcal{S}}$ rescales $\delta = \mathbf{x}' - \mathbf{x}$ if $\|\delta\|_{\ell_2} > \epsilon$ via $\delta \leftarrow \epsilon \cdot \delta / \|\delta\|_{\ell_2}$.

B OMITTED PROOFS & THEORETICAL RESULTS

To characterise how adversarial attacks exploit feature representations, we prove that the optimal perturbation weights each input dimension by the degree to which its corresponding feature aligns with the path to the decision boundary. The analysis concerns our linear models without ReLU activations, and it yields exact, closed-form expressions for both the optimal attack and the precise conditions under which attacks transfer.

Consider an input $\mathbf{x} \in \mathbb{R}^d$ encoded via $\phi(\mathbf{x}) = \mathbf{W}_e \mathbf{x}$ into the latent representation $\mathbf{h} \in \mathbb{R}^m$, where the columns of $\mathbf{W}_e \in \mathbb{R}^{m \times d}$ form an overcomplete dictionary $\{\mathbf{v}_i\}_{i=1}^d$ ($m < d$). For the argmax task we observe empirically that the trained encoder–decoder pair satisfies $\mathbf{W}_d = \mathbf{W}_e^{\top}$, so the logit for class j is $z_j = \mathbf{v}_j^{\top} \mathbf{h}$, with $\mathbf{v}_j$ the j -th column of $\mathbf{W}_e$, and the predicted class is $\hat{y} = \arg \max_j z_j$. Input perturbations map to latent perturbations as $\Delta \mathbf{h} = \mathbf{W}_e \delta$, where $\delta = (\delta_1, \dots, \delta_d)^{\top}$ collects the perturbation coefficients.

The decision boundary between classes j and k is the set of points at which their logits coincide, $\mathcal{B}_{jk} = \{\mathbf{h} : z_j = z_k\}$, equivalently $\{\mathbf{h} : (\mathbf{v}_k - \mathbf{v}_j)^{\top} \mathbf{h} = 0\}$. The vector $\mathbf{n} = (\mathbf{v}_k - \mathbf{v}_j)$ is the normal to this pairwise boundary, pointing in the direction of maximal increase of the margin $z_k - z_j$. We restate the propositions before giving their proofs.

Proposition 1 (Optimal targeted attack). The optimal input perturbations δ that maximise movement from class j toward class k under constraint $\|\delta\|_2 = \epsilon$ satisfy:

$$\delta \propto \mathbf{W}_e^{\top} \mathbf{n}$$

Proof. To move a sample from being classified as j to being classified as k , we need to maximise the logit margin $z_k - z_j$. Under perturbation δ , the new margin becomes:

$$z'_k - z'_j = \mathbf{v}_k^{\top} (\mathbf{h} + \Delta \mathbf{h}) - \mathbf{v}_j^{\top} (\mathbf{h} + \Delta \mathbf{h}) \quad (2)$$

$$= (\mathbf{v}_k - \mathbf{v}_j)^{\top} (\mathbf{h} + \mathbf{W}_e \delta) \quad (3)$$

$$= (\mathbf{v}_k - \mathbf{v}_j)^{\top} \mathbf{h} + (\mathbf{v}_k - \mathbf{v}_j)^{\top} \mathbf{W}_e \delta \quad (4)$$

The change in margin is $(\mathbf{v}_k - \mathbf{v}_j)^{\top} \mathbf{W}_e \delta = \delta^{\top} \mathbf{W}_e^{\top} (\mathbf{v}_k - \mathbf{v}_j)$. We seek:

$$\max_{\delta} \delta^{\top} \mathbf{W}_e^{\top} (\mathbf{v}_k - \mathbf{v}_j) \quad \text{subject to} \quad \|\delta\|_2 = \epsilon$$

Let $\mathbf{g} = \mathbf{W}_e^{\top} (\mathbf{v}_k - \mathbf{v}_j)$. By the Cauchy-Schwarz inequality:

$$|\delta^{\top} \mathbf{g}| \leq \|\delta\|_2 \|\mathbf{g}\|_2 = \epsilon \|\mathbf{g}\|_2$$

Equality is achieved when δ and $\mathbf{g}$ are parallel. Given the constraint $\|\delta\|_2 = \epsilon$:

$$\delta = \frac{\epsilon}{\|\mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)\|_2} \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$$

□

Corollary (Interference drives vulnerability). For a targeted attack from class j to class k , the adversarial perturbation magnitude for input feature i is:

$$|\delta_i| \propto |\mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j)|$$

where $\mathbf{v}_i$ is the i -th column of $\mathbf{W}_e$. This quantity represents the differential interference between feature i and the class representations.

This exposes exactly the mechanism by which superposition creates adversarial vulnerability. Each input feature i is perturbed in proportion to $\mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j)$ —the differential interference between feature i and the class representations. Under superposition the directions are non-orthogonal, so that even semantically unrelated features have non-zero inner products with $(\mathbf{v}_k - \mathbf{v}_j)$ and therefore constitute exploitable interference patterns. The multi-class setting compounds this vulnerability: with k classes there are many pairwise boundaries, each carrying its own interference pattern. Adversarial perturbations leverage these cross-feature dependencies, manipulating precisely the features that influence the decision through their interference with the class representations. This accounts exactly for the vulnerability observed empirically—attacks succeed not by directly increasing the target-class features, but by exploiting the web of interference that superposition necessarily creates.

Prop. 1 characterises the optimal perturbation direction for moving from class j to k , furnishing the gradient that maximises the margin $z_k - z_j$. When intervening classes exist (with $z_j < z_i < z_k$ for some class i), following this gradient may cause the model to predict i before it reaches the target k ; iterative methods such as PGD accommodate this by recomputing the gradient at each step, so that the global attack path emerges from repeated local decisions rather than from a single optimisation. Moreover, although our analysis assumes $\mathbf{W}_d = \mathbf{W}_e^\top$ on the basis of our empirical observations, the framework extends without modification to arbitrary decoders: for a general decoder $\mathbf{W}_d \in \mathbb{R}^{k \times m}$ with rows $\mathbf{w}_j^\top$, the optimal perturbation becomes $\delta \propto \mathbf{W}_e^\top (\mathbf{w}_k - \mathbf{w}_j)$, which captures the interference between the encoder directions and the decoder weights.

Proposition 2 (Attack transferability). Consider encoders ϕ and ψ with basis matrices $\mathbf{W}_e \in \mathbb{R}^{m \times d}$ and $\mathbf{W}'_e \in \mathbb{R}^{m \times d}$ whose columns are related by orthogonal transformation $\mathbf{v}'_i = \mathbf{Q}\mathbf{v}_i$ (where $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$). If both models use their encoder transpose as decoder (i.e., $\mathbf{W}_d = \mathbf{W}_e^\top$ and $\mathbf{W}'_d = (\mathbf{W}'_e)^\top$), then both models have identical optimal input perturbation vectors for any targeted attack from class j to class k .

Proof. For model ϕ , the optimal perturbation from Prop. 1 is:

$$\delta^\phi \propto \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$$

For model ψ with transformed columns $\mathbf{v}'_i = \mathbf{Q}\mathbf{v}_i$:

$$\delta^\psi \propto (\mathbf{W}'_e)^\top (\mathbf{v}'_k - \mathbf{v}'_j) = (\mathbf{W}'_e)^\top (\mathbf{Q}\mathbf{v}_k - \mathbf{Q}\mathbf{v}_j)$$

Since $\mathbf{W}'_e = \mathbf{Q}\mathbf{W}_e$ (all columns are transformed), we have:

$$\delta^\psi \propto (\mathbf{Q}\mathbf{W}_e)^\top \mathbf{Q}(\mathbf{v}_k - \mathbf{v}_j) \tag{5}$$

$$= \mathbf{W}_e^\top \mathbf{Q}^\top \mathbf{Q}(\mathbf{v}_k - \mathbf{v}_j) \tag{6}$$

$$= \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j) \tag{7}$$

where the last equality uses $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$. Thus δ^ϕ and δ^ψ have identical proportionality. Under the same norm constraint $\|\delta\|_2 = \epsilon$, we have $\delta^\phi = \delta^\psi$. □

Prop. 2 explains why attacks transfer between models trained under similar regimes. When two models learn feature representations that differ only by an orthogonal transformation—the same

geometric structure in a different orientation—they share identical vulnerability patterns in input space, precisely because an orthogonal transformation preserves every inner product between features and thereby preserves the interference patterns that attacks exploit. This accounts directly for our empirical finding that models trained on data with the same correlations exhibit high attack transferability: they discover the same feature geometry up to rotation, and hence inherit identical adversarial vulnerabilities.

C TOY MODEL EXPERIMENTS

This section provides supplementary details, extended results, and further discussion for the toy model experiments of the main paper. We report model accuracies across a considerably broader range of parameters than the main text, clarifying how model capacity and data characteristics such as sparsity jointly shape the learning process and delimit the conditions under which feature superposition arises. We then present additional visual examples that complement Fig. 1 and illustrate the mechanics of adversarial attacks under a variety of conditions.

C.1 HYPOTHESES TESTING FRAMEWORK

We now state explicitly the null and alternative hypotheses underlying the three research questions of Sec. 3, together with the controlled tests through which each is evaluated.

Research Q1: Do adversarial perturbations exploit superposition geometry?

- H_0 : Adversarial perturbations are random with respect to feature geometry.
- H_1 : Adversarial perturbations systematically exploit geometric relationships between superposed representations.

Research Q2: Do data correlations determine superposition geometry?

- H_0 : Input correlations have no systematic effect on learned geometries.
- H_1 : Input correlations determine geometric arrangements across model initializations.

Research Q3: Does shared geometry explain attack transferability?

- H_0 : Attack transferability is independent of geometric similarity.
- H_1 : Transferability increases with shared latent structure.

We test these hypotheses through controlled experiments:

- $H_1(1)$: we measure the alignment of the input perturbation profile with a class’s latent representation and with the latent attack vector.
- $H_1(2)$: we systematically vary the input correlations and measure the resulting geometries.
- $H_1(3)$: we quantify transferability rates across models of varying geometric similarity.

C.2 ACCURACY OF TOY MODEL FOR A RANGE OF PARAMETERS

The toy model experiments in the main paper predominantly used low-dimensional settings for conceptual clarity. To characterise the model’s behaviour comprehensively, this subsection reports the classification accuracies of the CE toy model as we vary the hidden-layer size (h), the number of classes (k), the number of features, and the sparsity level (S), thereby delineating the regime in which the models learn to represent features in superposition. The sparsity level S is the probability that a given input feature $x_i^{(j)}$ is set to zero, so that larger S denotes greater input sparsity; we report results as a function of the feature density $1 - S$, the probability that a feature is non-zero. Tab. 2 and Tab. 3 together map the model’s performance limits and its capacity to learn latent representations in superposition.

C.3 CORRELATIONS CONSTRAIN ARRANGEMENTS

Tab. 4 reports the results for *Do Input Correlations Determine Latent Feature Geometry?* across a broader range of class counts (k) and hidden dimensions (m), confirming that the strict ordering

Table 2: Classification accuracy of the CE toy model at a fixed bottleneck dimension ($m = 2$) across numbers of classes (k), total input features (features = $k \times 3$), and input feature densities ($1 - S$). The results reveal how input sparsity governs the performance degradation incurred as more classes must be superposed within a constrained latent space: at any fixed k , accuracy improves monotonically as density decreases, so that greater sparsity systematically permits more classes to be packed into the same two dimensions.

Classes (k)	Features	Hidden (m)	Accuracy at Input Feature Density ($1 - S$)							
			1.0	0.57	0.33	0.19	0.11	0.06	0.04	0.02
3	9	2	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
4	12	2	0.67	0.71	0.82	0.92	0.98	0.99	1.00	1.00
5	15	2	0.53	0.50	0.65	0.77	0.89	0.95	0.98	0.99
6	18	2	0.40	0.43	0.51	0.66	0.82	0.93	0.97	0.99
7	21	2	0.34	0.34	0.40	0.53	0.73	0.87	0.95	0.98
8	24	2	0.30	0.30	0.33	0.40	0.63	0.82	0.93	0.97
9	27	2	0.24	0.26	0.30	0.35	0.57	0.75	0.89	0.96
10	30	2	0.22	0.24	0.26	0.31	0.50	0.72	0.87	0.95
15	45	2	0.14	0.15	0.16	0.17	0.25	0.40	0.65	0.86
20	60	2	0.10	0.10	0.11	0.13	0.15	0.24	0.44	0.76
25	75	2	0.07	0.08	0.09	0.10	0.12	0.16	0.26	0.62
30	90	2	0.06	0.07	0.07	0.07	0.09	0.11	0.21	0.45

Table 3: Classification accuracy of the CE toy model across numbers of classes (k), total input features, bottleneck dimensions (m), and input feature sparsity levels (S), reported for several numbers of features per class (p). These larger-scale configurations confirm that the sparsity-dependent behaviour of Tab. 2 persists as model size grows, thereby establishing that the mechanism is in no way confined to the low-dimensional regime.

Classes (k)	Features	Hidden (m)	Accuracy at Input Feature Density ($1 - S$)							
			1.0	0.57	0.33	0.19	0.11	0.06	0.04	0.02
30	30	30	0.23	0.24	0.38	0.62	0.83	0.94	0.99	1.00
30	90	90	0.27	0.27	0.33	0.41	0.51	0.67	0.85	0.94
40	40	30	0.67	0.54	0.73	0.77	0.88	0.96	0.99	0.99
40	120	30	0.71	0.64	0.65	0.72	0.73	0.79	0.89	0.96
60	60	10	0.05	0.07	0.12	0.25	0.47	0.73	0.90	0.97
60	180	10	0.08	0.10	0.13	0.17	0.22	0.32	0.52	0.75
80	80	30	0.15	0.17	0.23	0.41	0.63	0.79	0.91	0.98
80	240	30	0.23	0.22	0.31	0.41	0.48	0.53	0.66	0.80
100	100	10	0.03	0.04	0.05	0.10	0.21	0.43	0.69	0.87
100	300	10	0.04	0.05	0.06	0.09	0.12	0.15	0.25	0.45

uncorrelated $<$ paired $<$ global holds across every configuration tested and is therefore not an artefact of the specific (k, m) used in the main text but a general property of the correlation structure.

Table 4: Geometric similarity across correlation types and superposition configurations. For every (k, m) configuration, geometric similarity increases monotonically from uncorrelated to paired to global correlations, reproducing the main-text result across scales.

Correlation Type	k	m	Geometric Similarity
Uncorrelated	6	2	0.18 ± 0.15
	30	10	0.17 ± 0.02
	90	30	0.17 ± 0.01
Paired	6	2	0.47 ± 0.07
	30	10	0.26 ± 0.07
	90	30	0.25 ± 0.01
Global	6	2	0.92 ± 0.04
	30	10	0.88 ± 0.01
	90	30	0.80 ± 0.01

C.4 HOW THE TOY MODEL MAPS TO REAL-WORLD MODELS

How does scaling networks affect adversarial vulnerability? Although empirical evidence indicates that larger models tend to be somewhat more robust, the effect is weak; under adversarial training clearer scaling trends emerge, yet the improvements remain largely specific to the attack type used during training rather than conferring general robustness (Howe et al., 2025).

Two competing phenomena operate as models scale. Larger models possess greater capacity to represent concepts, but general tasks such as next-token prediction over internet text present a long tail of useful concepts to capture. Consequently, superposition remains prevalent even in frontier models despite their increased capacity (Lindsey et al., 2025). Dictionary-learning methods corroborate this directly: SAEs require progressively larger dictionaries for larger models (Gao et al., 2025), indicating that the number of represented features scales with model size. The fundamental tension that drives superposition—the need to compress many features into limited dimensions—therefore does not dissolve with scale but persists.

Because both superposition and adversarial vulnerability persist in large-scale models, the mechanism we identify remains relevant across the full range of model scales rather than being confined to small networks. Understanding how the geometry of superposition evolves with scale, and how this evolution might be leveraged to mitigate vulnerability, is a compelling direction for future work.

C.4.1 SEPARATING SUPERPOSITION EFFECTS FROM CAPACITY REDUCTION

Holding the bottleneck dimension fixed cleanly disentangles the otherwise confounded effects of superposition pressure and capacity reduction. Fixing $m = 2$ and varying the number of classes k isolates superposition pressure in isolation; we report the resulting model accuracy, robust accuracy, and transferability in Tab. 5. These controls establish decisively that it is increased superposition pressure (k/m), and not reduced capacity, that drives the adversarial vulnerabilities we observe.

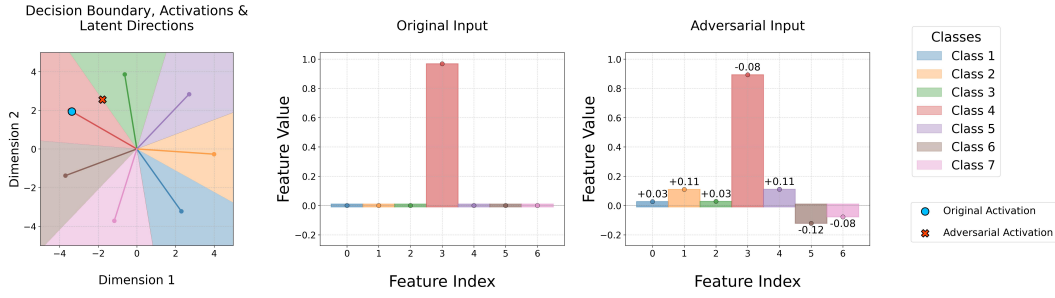
C.4.2 ADDITIONAL EXAMPLES OF AEXS IN TOY MODEL

Fig. 1 illustrates how adversarial attacks exploit the interference between latent features in superposition. Here we provide further visual examples (Fig. 5) that corroborate this mechanism across a range of varied conditions, supplementing the main text with:

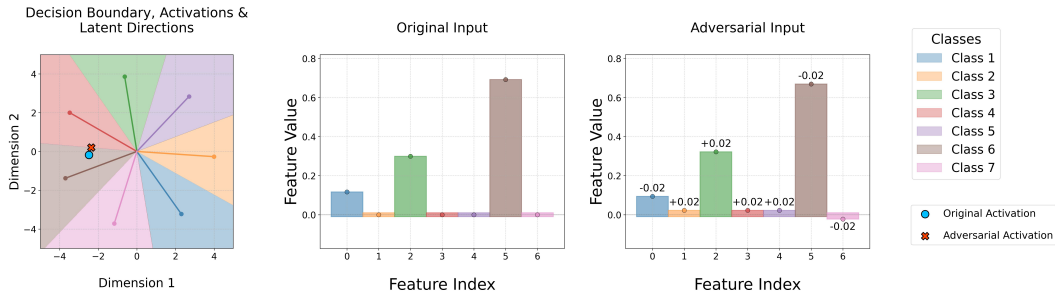
- Fig. 5a: a further instance of the setup in Sec. 3.1 ($m = 2, k = 7$) under an ℓ_2 -norm PGD attack, showing the IPP together with the latent-space manipulations that produce misclassification.
- Fig. 5b: a comparable setup ($m = 2, k = 7$) under an ℓ_∞ -norm PGD attack applied to a less sparse input.

Table 5: Performance metrics across correlation types, perturbation budgets (ε), and numbers of classes (k). With the bottleneck dimension held fixed, robust accuracy falls as k (and hence superposition pressure) grows, while attack transferability remains consistently far higher under full correlation than under no correlation—jointly isolating superposition, rather than capacity, as the driver of vulnerability.

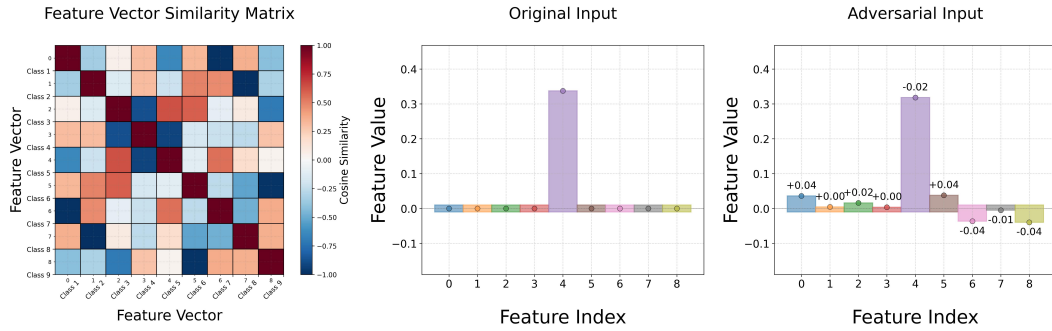
Correlation Type	Parameters		Performance (%)		
	ε	k	Accuracy	Robust Acc.	Attack Trans.
Uncorrelated	0.05	2	100.0	100.0	0.0
		4	100.0	94.9	21.2
		6	99.9	87.0	9.8
		8	99.8	77.7	17.5
	0.1	2	100.0	100.0	0.0
		4	100.0	88.1	9.8
		6	99.7	75.4	9.9
		8	99.6	56.3	23.5
	0.5	2	100.0	100.0	0.0
		4	100.0	99.4	0.0
		6	99.9	62.9	30.5
		8	99.5	24.9	39.0
Fully Correlated	0.05	2	100.0	100.0	0.0
		4	100.0	95.8	98.4
		6	97.3	70.6	98.4
		8	93.0	57.4	97.2
	0.1	2	100.0	100.0	0.0
		4	100.0	88.7	98.5
		6	97.7	45.6	99.7
		8	94.2	34.8	98.6
	0.5	2	100.0	100.0	0.0
		4	100.0	90.0	100.0
		6	97.2	34.6	100.0
		8	93.0	32.0	100.0



(a) An ℓ_2 -norm attack that flips the classification of a class-4 input to class 3. The left plot shows the original and adversarial activations in latent space together with the representation directions; the right plots show the original and perturbed input feature values, whose signs and magnitudes again track the latent geometry.



(b) An ℓ_∞ -norm attack that flips the classification of a class-6 input to class 4. The left plot shows the original and adversarial activations in latent space relative to the class latent directions; the right plots show the original and perturbed input feature values, confirming that the mechanism persists under the ℓ_∞ norm and at lower input sparsity.



(c) An ℓ_2 -norm adversarial attack in a 7-class setup with an increased bottleneck dimension $m = 3$. The leftmost plot now shows the cosine-similarity matrix between the latent representations of each pair of classes, making the interference structure that the attack exploits directly visible.

Figure 5: Visualisations of AExs in the toy model, supplementing Figure 1 of the main paper by illustrating the attack mechanism in both activation space and input space under varied norms, sparsities, and bottleneck dimensions. Across every condition the perturbation is fixed by the latent geometry.

- Fig. 5c: an example at increased bottleneck dimensionality ($m = 3$, $k = 7$) under an ℓ_2 -norm PGD attack, additionally displaying the feature-vector similarity matrix used to compute geometric similarity.

D CIFAR-10 EXPERIMENTS

To establish that the principles observed in the toy models extend to substantially more complex settings, Sec. 4 introduces experiments on CIFAR-10 (Krizhevsky, 2009) with an engineered bottleneck. This section details the setup and presents the full set of extended results.

D.1 ARCHITECTURE & TRAINING INFORMATION

The base ViT (Dosovitskiy et al., 2020) comprised 6 transformer layers, an embedding dimension (d) of 512, and 8 attention heads per layer. The 32×32 -pixel CIFAR-10 images were partitioned into 4×4 -pixel patches, the Multilayer Perceptron (MLP) within each transformer block had a hidden dimension of 512, and learned positional embeddings were used.

The bottleneck architecture consisted of a linear encoder followed by a linear decoder: the encoder projected the pre-classification activations of the ViT backbone into an m -dimensional latent space ($m \in \{2, 3, 5, 10\}$), and the decoder mapped these m -dimensional representations back to the $k = 10$ CIFAR-10 classes.

The base ViT was trained on CIFAR-10 for 250 epochs with the Adam optimiser (default PyTorch parameters) at a learning rate of 0.001, a batch size of 512, a cosine-annealing learning-rate schedule, CE loss, and dropout, across five random seeds to account for variability. After training, the base ViT weights were frozen and the bottleneck layer was trained for 30 epochs at a learning rate of 0.001. For preprocessing, images undergo RandomCrop with 4-pixel padding, a resize to the target size (32×32 by default), and RandomHorizontalFlip for data augmentation.

D.1.1 NORMALISED ROBUST ACCURACY ACROSS PERTURBATION MAGNITUDES

Fig. 4 (right) reports how normalised robust accuracy varies with bottleneck dimension at a single value of ϵ . This subsection generalises that finding by reporting normalised robust accuracy under PGD attacks of varying strength, for both ℓ_2 -norm (Tab. 7) and ℓ_∞ -norm (Tab. 6) attacks, thereby furnishing a comprehensive picture of how the degree of superposition and the attack strength jointly determine robustness.

AExs for these evaluations were generated with PGD using 100 iterations and a step size (α) of 0.01. Robust accuracy was evaluated on 500 samples per configuration, and we report the mean normalised robust accuracy together with its standard deviation across five random seeds.

Table 6: Mean normalised robust accuracy ($\pm$ standard deviation across 5 seeds) for ViT models with different bottleneck dimensions (m) on CIFAR-10 under ℓ_∞ -norm PGD attacks of varying perturbation magnitude (ϵ), normalised by each bottlenecked model’s clean accuracy. Consistent with Sec. 4, at every ϵ robust accuracy increases monotonically with m —equivalently, robustness declines as superposition intensifies—confirming that the trend holds across the full range of attack strengths.

ϵ	Bottleneck Dimension (m)			
	2	3	5	10
0.001	96.0% $\pm$ 0.4%	97.4% $\pm$ 0.7%	97.9% $\pm$ 0.4%	98.1% $\pm$ 0.3%
0.01	61.7% $\pm$ 3.6%	69.6% $\pm$ 3.5%	77.0% $\pm$ 0.9%	81.8% $\pm$ 3.2%
0.05	4.9% $\pm$ 1.5%	6.7% $\pm$ 1.6%	9.3% $\pm$ 0.6%	10.5% $\pm$ 0.8%
0.1	0.1% $\pm$ 0.2%	0.2% $\pm$ 0.3%	0.2% $\pm$ 0.4%	0.2% $\pm$ 0.4%
0.5	0.0% $\pm$ 0.0%	0.0% $\pm$ 0.0%	0.0% $\pm$ 0.0%	0.0% $\pm$ 0.0%

D.1.2 ATTACK TRANSFERABILITY ACROSS PERTURBATION MAGNITUDES

We report attack transferability across perturbation magnitudes (ϵ) and bottleneck dimensions (m): Tab. 9 presents the ℓ_2 -norm transferability and Tab. 8 the ℓ_∞ -norm transferability.

Table 7: Mean normalised robust accuracy ($\pm$ standard deviation across 5 seeds) for ViT models with different bottleneck dimensions (m) on CIFAR-10 under ℓ_2 -norm PGD attacks of varying perturbation magnitude (ϵ), normalised by each bottlenecked model’s clean accuracy. Consistent with Sec. 4, robustness decreases monotonically with smaller m (greater superposition) and with larger ϵ , mirroring the ℓ_∞ results and confirming that the trend is norm-independent.

ϵ	Bottleneck Dimension (m)			
	2	3	5	10
0.1	90.4% $\pm$ 1.7%	91.8% $\pm$ 0.7%	94.6% $\pm$ 1.2%	95.0% $\pm$ 0.7%
0.5	58.5% $\pm$ 5.0%	60.8% $\pm$ 5.0%	69.2% $\pm$ 1.8%	72.6% $\pm$ 2.8%
1.0	41.7% $\pm$ 4.8%	44.0% $\pm$ 3.0%	50.4% $\pm$ 0.6%	54.9% $\pm$ 1.7%
2.0	34.0% $\pm$ 4.6%	36.2% $\pm$ 3.3%	41.5% $\pm$ 1.3%	47.4% $\pm$ 2.4%
5.0	30.2% $\pm$ 4.1%	32.9% $\pm$ 3.5%	37.3% $\pm$ 1.9%	42.7% $\pm$ 1.0%

D.2 CORRELATIONS IN CLASS FEATURES

In Sec. 3 it was the correlations between inputs that drove the superposition arrangement. Here, by contrast, the arrangement is driven not by correlations between input classes but by correlations among the representations present at this point in the network. At the classification layer these correlations plausibly resemble the confusion matrix—the pattern of how each class is misclassified relative to the others—so that the classes clustering together are exactly those the network finds inherently similar and misclassifies together. To test this interpretation, we repeat the experiment while fine-tuning the base ViT from timm/vit_base_patch16_384, pretrained on ImageNet-21k (14 million images, 21,843 classes) and ImageNet (1 million images, 1,000 classes). After fine-tuning on CIFAR-10 and applying the same bottleneck training procedure as Sec. 4, the resulting geometry exhibits random ordering across initialisations with approximately equal spacing between features (*i.e.* neural collapse Kothapalli (2023)); performance is near 100%, so that the confusion matrix is effectively the identity and the superposed geometry loses its structure. Models trained solely on CIFAR-10, by contrast, converge to similar geometries precisely because they share the same learned difficulty structure—the same pairs of classes prove hard to distinguish—which in turn yields consistent superposition patterns. This control confirms that it is the correlation structure, and not the bottleneck alone, that determines the geometry.

D.2.1 RESNET-18

To confirm that our findings generalise across architectures, we repeat the experiments with a ResNet-18 (He et al., 2016) in place of the ViT as the base model. This yields a slightly higher clean accuracy of 92% (versus 89% for the ViT), and robust accuracy falls marginally faster for ResNet-18 at matched ϵ . The same qualitative trends nonetheless hold intact: normalised robust accuracy declines and transferability rises as bottleneck pressure increases. That the mechanism reproduces on a convolutional architecture just as it does on a transformer establishes that it is a property of superposition itself rather than of any particular architecture.

E MODULAR ADDITION

Sec. 5 introduces a modular addition task to probe adversarial vulnerability in a setting where features are represented orthogonally. This section details the experimental setup, provides additional results, and expands on why we regard this form of representation-level analysis as a broadly applicable instrument.

E.1 LEARNED ALGORITHMIC REPRESENTATION IN MODULAR ADDITION

As established by Nanda et al. (2023) and Prieto et al. (2025b), the network discovers and implements a trigonometric algorithm for this task: the encoder projects inputs onto non-basis-aligned linear representations encoding $\sin(\omega_k a)$ and $\cos(\omega_k a)$ at the key frequencies $\omega_k = 2\pi k/P$. These features

Table 8: Attack transferability (%) for ℓ_∞ -norm PGD attacks on CIFAR-10 ViT models. Each row reports transferability from a model trained with a given ‘Source Seed’ (e.g. Seed 10) to three target models, each trained with one of the seeds in the sub-header (e.g. ‘vs. Seeds 20/30/40’); the three slash-separated values in a cell give the transferability to these three target seeds in order. All models within a row share the same bottleneck dimension m , and the ‘Mean $\pm$ Std’ column averages transferability over all 12 source–target seed pairings for each (ϵ, m) configuration. Consistent with Sec. 4, transferability is systematically higher at smaller m , confirming that greater superposition yields more consistent latent geometries and hence higher transferability across the full range of budgets.

ϵ	m	Seed 10 vs. Seeds 20/30/40	Seed 20 vs. Seeds 10/30/40	Seed 30 vs. Seeds 10/20/40	Seed 40 vs. Seeds 10/20/30	Mean $\pm$ Std
0.001	2	77.8/66.7/44.4	25.0/62.5/25.0	72.7/72.7/72.7	58.3/75.0/58.3	59.3 ± 17.7
	3	57.1/28.6/28.6	14.3/28.6/28.6	63.6/72.7/72.7	42.9/42.9/42.9	43.6 ± 18.4
	5	70.0/50.0/50.0	57.1/71.4/57.1	50.0/25.0/25.0	16.7/50.0/33.3	46.3 ± 16.9
	10	55.6/55.6/66.7	12.5/37.5/25.0	16.7/33.3/16.7	28.6/71.4/28.6	37.3 ± 19.3
0.01	2	60.3/52.6/48.7	63.8/42.0/60.9	53.8/40.9/52.7	51.2/55.8/46.5	52.4 ± 6.9
	3	42.9/46.4/50.0	46.4/42.9/49.1	44.2/38.9/44.2	42.9/45.1/41.8	44.6 ± 3.0
	5	43.3/41.1/43.3	39.2/30.4/41.8	46.0/40.2/47.1	41.2/35.3/32.9	40.2 ± 4.8
	10	42.5/46.0/44.8	38.0/39.4/46.5	43.5/47.8/47.8	32.6/40.0/35.8	42.1 ± 4.7
0.05	2	35.0/36.4/37.8	38.8/36.0/46.3	37.4/31.3/41.7	42.0/42.0/40.7	38.8 ± 3.8
	3	29.9/29.9/37.1	30.4/30.1/35.6	30.2/30.2/35.1	30.1/31.9/34.4	32.1 ± 2.6
	5	26.8/25.3/29.2	25.2/21.5/26.4	28.0/28.0/26.5	26.0/26.6/21.9	25.9 ± 2.2
	10	27.1/24.9/30.4	21.6/23.0/28.4	25.3/28.1/30.6	21.0/24.9/20.1	25.4 ± 3.4
0.1	2	34.2/35.6/38.2	36.7/34.9/44.5	37.9/30.0/40.8	41.2/42.0/40.3	38.0 ± 3.8
	3	29.6/28.9/35.2	28.6/29.2/34.8	29.1/28.2/34.0	28.8/31.1/32.8	30.9 ± 2.5
	5	25.4/23.3/27.0	23.8/19.6/24.6	25.9/26.7/25.4	24.6/24.3/20.7	24.3 ± 2.1
	10	24.8/22.6/29.7	20.1/21.6/26.1	22.5/25.5/28.0	19.5/23.3/18.7	23.5 ± 3.3

are represented orthogonally—not in superposition—and their sinusoidal values are recoverable by linear probes. The MLP then applies sum-angle identities to compute $\cos((a+b)\omega_k)$ for various k , and these terms interfere constructively at the logits so as to select the correct class. AExs are generated by relaxing the one-hot constraint and perturbing the input $\mathbf{x}_j \in \mathbb{R}^P$ (encoding integer j) to $\mathbf{x}' = \mathbf{x} + \delta$ via ℓ_2 - and ℓ_∞ -norm PGD; for simplicity only a is attacked while b is held fixed.

As explained by Nanda et al. (2023), the network learns a structured trigonometric algorithm in which W_e encodes each integer’s sine and cosine values at the key frequencies. Because these are linearly decodable sinusoidal values, scaling the corresponding feature vector alters the encoded sinusoidal value itself rather than merely its activation intensity—and it is exactly this property that the attack exploits. The integer embeddings W_e^j project onto a Fourier-like basis, inducing a circular, periodic structure in the embedding space; unlike the arrangements produced by data correlations, this circularity arises directly from the sine/cosine encoding of the (uncorrelated) input integers.

Our model learns to use approximately seven key frequencies (ω_k). Each embedding W_e^j for an input integer x_j predominantly encodes its fourteen corresponding trigonometric features, $\cos(\omega_i x_j)$ and $\sin(\omega_i x_j)$ for $i \in \{1, 2, \dots, 7\}$, so that the concatenated 200-dimensional vector supplies roughly 28 key trigonometric features (fourteen for a and fourteen for b) to the MLP. The MLP then composes these, via sum-angle identities (e.g. $\cos(A+B) = \cos A \cos B - \sin A \sin B$), to compute $(x_1 + x_2) \pmod{P}$. The algorithmic brittleness studied in Sec. 5 stems precisely from adversarial perturbation of these key learned trigonometric features.

Table 9: Attack transferability (%) for ℓ_2 -norm PGD attacks on CIFAR-10 ViT models. The format, detailing source-to-target transferability, mirrors Tab. 8 (see its caption for a full explanation). These ℓ_2 results independently confirm the claim of Sec. 4 that greater superposition yields higher attack transferability, establishing that the effect is not specific to the ℓ_∞ norm.

ϵ	m	Seed 10 vs. Seeds 20/30/40	Seed 20 vs. Seeds 10/30/40	Seed 30 vs. Seeds 10/20/40	Seed 40 vs. Seeds 10/20/30	Mean $\pm$ Std
0.1	2	70.6/64.7/41.2	60.0/50.0/60.0	64.3/60.7/57.1	48.1/63.0/51.9	57.6 ± 8.0
	3	39.1/39.1/47.8	50.0/60.7/50.0	48.1/55.6/51.9	48.0/28.0/40.0	46.5 ± 8.4
	5	43.5/34.8/43.5	31.6/31.6/42.1	42.9/35.7/21.4	30.4/39.1/34.8	35.9 ± 6.4
	10	45.5/59.1/45.5	19.0/28.6/28.6	37.5/43.8/43.8	23.8/52.4/38.1	38.8 ± 11.4
0.5	2	59.6/52.1/48.9	60.8/35.4/60.8	54.1/39.4/51.4	43.8/49.5/46.7	50.2 ± 7.7
	3	38.7/40.6/48.1	36.1/36.8/41.7	37.4/36.5/43.5	34.4/34.4/36.7	38.7 ± 3.9
	5	38.4/32.0/37.6	29.4/25.7/32.1	39.1/33.9/38.3	31.8/29.9/32.7	33.4 ± 4.0
	10	38.3/37.4/37.4	26.5/28.4/31.4	32.4/35.3/35.3	23.4/28.2/25.0	31.6 ± 5.0
1.0	2	50.8/43.7/44.4	51.7/36.4/50.0	46.1/32.9/44.7	42.7/46.7/45.3	44.6 ± 5.3
	3	35.7/36.9/41.7	31.8/36.4/42.6	33.7/32.0/37.3	34.9/32.0/35.5	35.9 ± 3.3
	5	32.1/28.9/33.2	30.9/23.0/30.9	35.9/31.5/33.7	27.7/28.2/27.1	30.3 ± 3.3
	10	32.3/32.3/38.5	23.7/27.7/29.9	29.1/32.0/34.3	22.2/25.6/21.1	29.1 ± 5.0
2.0	2	47.0/41.6/44.3	48.2/36.2/48.2	43.9/33.5/42.1	43.4/43.4/42.8	42.9 ± 4.2
	3	31.9/37.7/40.3	31.1/35.6/40.6	32.6/31.1/36.8	35.4/33.3/34.3	35.1 ± 3.2
	5	29.9/25.4/32.1	26.3/22.1/26.7	32.3/31.8/29.5	27.9/27.9/23.6	28.0 ± 3.2
	10	30.8/30.4/35.7	22.5/24.9/28.6	24.8/29.7/32.7	20.4/26.1/19.0	27.1 ± 4.9
5.0	2	44.7/46.7/40.8	45.3/37.3/48.7	42.7/33.9/41.5	42.3/42.3/40.6	42.2 ± 3.8
	3	32.3/35.8/40.8	31.3/32.6/39.9	31.0/30.0/33.8	35.0/33.0/33.0	34.1 ± 3.2
	5	30.6/24.1/28.6	28.3/20.9/24.8	31.7/31.3/27.4	28.1/29.0/23.2	27.3 ± 3.3
	10	29.8/27.4/32.7	19.7/24.1/28.9	24.9/29.9/31.2	18.8/25.8/19.2	26.0 ± 4.6

E.2 EXPERIMENT DETAILS FOR MODULAR ADDITION TASK

The network is trained for addition modulo 113. Inputs are integer pairs (a, b) with $a, b \in \{0, \dots, 112\}$, and each integer is mapped to a 100-dimensional vector via a shared embedding $\mathbf{W}_E \in \mathbb{R}^{113 \times 100}$. The embeddings $\mathbf{W}_E^a$ and $\mathbf{W}_E^b$ are concatenated into a 200-dimensional feature vector that feeds a 3-hidden-layer MLP (200 ReLU neurons per layer), whose output layer produces 113 classification logits. The model is trained end-to-end with cross-entropy loss and AdamW (weight decay 4), reaching 100% test accuracy. AExs are generated with PGD under the ℓ_2 and ℓ_∞ norms at varying ϵ ; each attack perturbs the embedding $\mathbf{W}_E^a$ (with integer b held fixed) by adding δ so as to induce misclassification of $(a + b) \pmod{113}$.

E.3 ALGORITHMIC BRITTLINESS

The frequency-aligned perturbations induce misclassification by corrupting the sinusoidal values read off from the encoder features. The hypothesised neuron-aligned case provides the clearest intuition: $\mathbf{W}_e$ would contain a row for the sine or cosine at each ω_k , whose entry in column $\mathbf{w}_e^a$ holds the value $\sin(\omega_k a)$ or $\cos(\omega_k a)$. For the input a , the one-hot $\mathbf{x}_a$ selects $\mathbf{w}_e^a$, giving $\mathbf{h}_a = \mathbf{w}_e^a$. The perturbation δ then transforms the one-hot $\mathbf{x}_a$ into a dense vector $\mathbf{x}'_a$, so that $\mathbf{h}'_a = \mathbf{W}_e \mathbf{x}'_a = (1 + \epsilon_a) \mathbf{h}_a + \sum_{k \neq a} \epsilon_k \mathbf{w}_e^k$. The contributions introduced by the $k \neq a$ sinusoidal terms corrupt the sinusoidal values, thereby disrupting the MLP’s downstream computation and inducing misclassification. Because the algorithm relies on the constructive interference of every ω_k

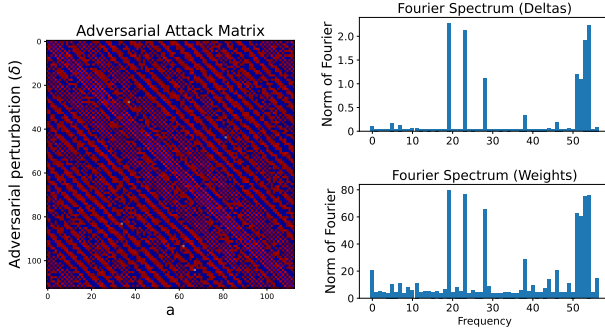


Figure 6: Left: each column is the mean attack δ added to a to successfully attack $a + b \bmod p$. Right: the norms of the Fourier components of δ and of $W_{MLP_{out}}$, demonstrating that the perturbation’s spectral energy concentrates precisely at the model’s key frequencies—the direct and unambiguous signature of the frequency-based attack mechanism.

at the logits, disrupting a single ω_k is insufficient—the remaining components still overwhelmingly support the correct answer—so that an effective attack must instead find a δ that maximally alters multiple frequency components at once.

E.4 ANALYTIC CONSTRUCTION OF ATTACKS AND BASELINE COMPARISONS

The main paper shows how understanding the learned trigonometric algorithm in the modular addition task enables the analytic, gradient-free construction of adversarial attacks. Sec. 5 benchmarks this ‘informed attack’ against PGD and a set of naive baselines; we specify those baseline methods precisely here. The baselines are:

- **PGD Attack:** standard PGD used to find the adversary.
- **Informed Attack:** the analytically constructed perturbation, built from combinations of cosines.
- **Single a :** the minimal perturbation of a single input entry sufficient to cause misclassification.
- **ℓ_∞ Noise:** random noise δ_{noise} added to the input a , with each component sampled independently and uniformly from $[-\epsilon, \epsilon]$; we report the smallest ϵ that induces a misclassification.
- **Uniform δ :** a perturbation with all components equal, $\delta_j = c \cdot \mathbf{1}$ for a scalar c ; we report the magnitude of c for which adding δ causes a misclassification.

E.5 DETAILS ON CERTIFIED ROBUST TRAINING

To assess the impact of defences on the identified vulnerability, we performed certified robust training aimed at rendering the network provably robust against ℓ_∞ -norm perturbations. We employed the abstraction-based technique (Wong & Kolter, 2018; Raghunathan et al., 2018; Palma et al., 2022) RSIP-IBP (Zhang et al., 2020; Henriksen & Lomuscio, 2023), which combines Interval Bound Propagation (IBP) (Katz et al., 2017; Ehlers, 2017) with Reversed Symbolic Interval Propagation (RSIP) to compute reachable output sets for the network layers and thereby formally certify robustness within a given perturbation budget. The training loss combined a standard loss component with an RSIP loss component, and certified training ran for 1000 epochs—the first 200 serving as a warm-up phase—at a learning rate of 1×10^{-3} and weight decay 1×10^{-4} , using the same train/test splits as the standard model. The results across training perturbation budgets (ϵ_{train}) and attack perturbation budgets (ϵ_{attack}) are summarised in Tab. 11.

As discussed in Sec. 5, when these robustly trained models were subsequently attacked with PGD, the character of the successful perturbations remained unchanged: they continued to exploit the same key frequency components inherent to the model’s learned algorithm. The sole effect of certified training was to increase the perturbation magnitude required for misclassification at the epsilons against which the model was successfully certified; it left the attack vector’s alignment with the model’s

Table 10: Minimum ℓ_2 -norm budget required to successfully attack $0 + 0 \bmod p$ across attack types. The informed, gradient-free attack is nearly as efficient as gradient-based PGD and dramatically more efficient than every naive baseline, confirming that it targets exactly the correct mechanism.

Attack Method	Success ϵ
PGD	0.17
Informed Attack	0.22
Single a	0.97
ℓ_∞ Noise	2.04
Uniform δ	6.20

Table 11: Verified robust accuracy (%) for modular addition models under ℓ_∞ -norm perturbations, comparing a standard (pre-robustification) model with models certifiably trained via RSIP-IBP at different training budgets (ϵ_{train}) and evaluated against various attack budgets (ϵ_{attack}). Consistent with Sec. 5, certified training raises robustness up to the specified ϵ_{train} , yet attacks continue to succeed at higher ϵ_{attack} by exploiting the same algorithmic sensitivities—demonstrating that certification shifts, rather than removes, the failure mode.

Attack Epsilon (ϵ_{attack})	Train Epsilon (ϵ_{train})	Standard Acc. (Pre-Rob., %)	Robust Acc. (Pre-Rob., %)	Standard Acc. (Robust, %)	Robust Acc. (Robust, %)
5.0×10^{-4}	10^{-4}	100.0	100.0	100.0	100.0
	10^{-3}				100.0
1.0×10^{-3}	10^{-4}	100.0	76.0	100.0	93.8
	10^{-3}				100.0
2.0×10^{-3}	10^{-4}	100.0	0.0	100.0	0.0
	10^{-3}				95.7
5.0×10^{-3}	10^{-4}	100.0	0.0	100.0	0.0
	10^{-3}				85.3

algorithmic sensitivities entirely intact. This confirms decisively that norm-bounded certification treats the symptom rather than the cause.

E.6 ON A POTENTIAL RECONCILIATION WITH SUPERPOSITION VULNERABILITY

Although the vulnerability studied here arises in the absence of feature superposition, it bears a striking phenomenological resemblance to superposition-driven vulnerabilities through a shared dependence on sparsity in the input representation. This resemblance motivates our *reconciliation conjecture*: algorithmic vulnerabilities may persist in networks with orthogonal representations whenever sparsity assumptions are violated upstream. Should earlier layers produce noisy or non-sparse features—potentially through adversarial interference—these deviations can propagate into otherwise well-isolated downstream circuits, so that a circuit which operates correctly on clean inputs nonetheless fails because the input it receives is effectively already perturbed. This furnishes a plausible route by which algorithm-disruption mechanisms such as the frequency attacks studied here could be triggered even in the absence of superposition, and it thereby unifies the two mechanisms under a single sparsity-dependent account. The interaction between upstream superposition and downstream algorithmic brittleness—and, more broadly, a layer-by-layer account of the specific vulnerability mechanisms that AExs trigger—remains underexplored; we leave a detailed investigation, grounded in both empirical evidence and theoretical modelling, as a promising direction for future work.

F INSPECTING ADVERSARIAL PERTURBATIONS VIA SAE FEATURES

The analyses of the main paper concerned settings with either explicitly defined class features (the toy model) or class-level representations in a bottleneck (the ViT experiments). In large, practical networks, by contrast, interference occurs among unlabelled latent features distributed across all layers. As noted in the future-work discussion, SAEs offer a promising unsupervised means of extracting linear features from NNs (Bricken et al., 2023). This section presents a preliminary investigation into using SAE-extracted features to characterise how a large-scale ViT responds internally to AExs relative to clean inputs. These initial results are intended to establish the feasibility of the approach for future work on understanding adversarial phenomena at the feature level in large models, rather than to advance conclusive new findings in their own right. We first outline the experimental setup and validate the SAEs’ behaviour on adversarial inputs, then present exploratory layer-wise feature-difference analyses.

F.1 EXPERIMENT SETUP

We analyse SAE-extracted features from the vision encoder of the LAION/CLIP-ViT-B-32-DataComp.XL-s13B-b90K model, a 12-layer ViT with $\approx 70\%$ zero-shot accuracy on ImageNet-1k Deng et al. (2009). We use the open-source Prisma SAEs, which map the 768-dimensional activation space to a dictionary of 49,152 features ($64\times$ expansion) (Joseph et al., 2025), examining SAEs trained on either mean patch-token or CLS-token representations across all layers. We analyse features activating above a high threshold (0.1), which isolates changes to the most strongly activating features, and above a low threshold (0.001). For the feature comparisons we generate 10,000 pairs of original and successfully attacked images (ℓ_∞ -norm, $\epsilon = 0.02$). This setup supports an initial, controlled exploration of how adversarial perturbations manifest at the level of SAE features.

F.2 SAE VALIDITY FOR ADVERSARIAL INPUTS

Because AExs are out-of-distribution inputs whereas the SAEs are trained conventionally, we first conduct a basic validation. This subsection examines the effect on model accuracy and SAE reconstruction of inserting the SAEs at inference time for both clean and adversarial inputs, thereby verifying that the SAEs process features relevant to the classification task even under adversarial conditions—an essential prerequisite for interpreting the subsequent differences in feature activations.

Table 12: Impact on ImageNet-1k classification accuracy of inserting a pre-trained SAE (one layer at a time) at inference for original, ℓ_∞ adversarial ($\epsilon = 0.02$), and random noisy inputs. ‘Accuracy’ denotes the ViT’s top-1 accuracy, and ‘Original input + SAE’ means the clean ViT activations at a given layer are passed through the SAE and its reconstruction forwarded to the next layer. The slight ‘de-attacking’ effect (higher accuracy for ‘Adv. input + SAE’ than for ‘Adv. input’) indicates that the SAEs filter some adversarial noise; the model nonetheless still overwhelmingly misclassifies, which confirms that the salient adversarial changes responsible for misclassification are preserved and processed by the SAE.

Metric	Layer 2	Layer 5	Layer 8	Layer 11
Accuracy				
Original input	1.00	1.00	1.00	1.00
Original input + SAE	0.99	0.99	0.99	0.86
Adv. input	0.00	0.00	0.00	0.00
Adv. input + SAE	0.02	0.01	0.01	0.11
Noisy input	0.98	0.98	0.98	0.98
Noisy input + SAE	0.98	0.98	0.98	0.85
SAE Improvement				
Original input	-0.01	-0.01	-0.01	-0.14
Adv. input	0.02	0.01	0.01	0.11
Noisy input	-0.00	-0.00	-0.00	-0.13

F.3 LAYER-WISE SAE FEATURE DIFFERENCES

This subsection presents initial exploratory visualisations comparing SAE feature activations across ViT layers for original, ℓ_∞ adversarial, and random noisy inputs. These visualisations are intended to identify avenues for future, more detailed investigation into how adversarial attacks distinctively reshape sparse feature representations. Concretely, we examine:

- Activation count: histograms comparing the number of SAE features activated above a set threshold in a layer for original, adversarial, and noisy images, probing whether attacks characteristically alter feature sparsity.
- Feature overlap: the degree of overlap (number of jointly activated features) between the original-vs.-adversarial and original-vs.-noisy conditions, probing whether attacks activate a distinctly different feature set.

- Jaccard similarity: the Jaccard similarity between sets of activated features, a normalised measure of overlap.
- Mean activation: histograms of mean activation values for features active above a threshold, probing changes in the intensity of (key) features.
- Distinct features: the count of unique features activated across an image set for each condition.

F.4 CLS TOKEN SAE FEATURES WITH HIGH ACTIVATION THRESHOLD (0.1)

This subsection presents a preliminary analysis of SAE features derived from CLS tokens, restricted to features activated above a threshold of 0.1. Figures 7 through 10 visualise, respectively, the histogram of activation counts, the feature overlap, the Jaccard similarity, and the histogram of mean activations for original, adversarial, and random noisy inputs across the ViT layers.

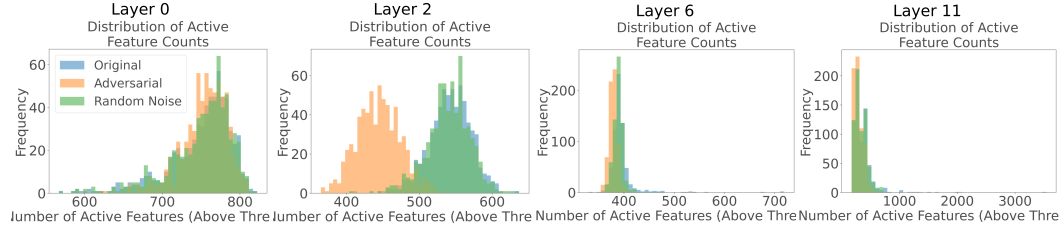


Figure 7: Histogram of activation counts for CLS token SAE features (threshold 0.1).

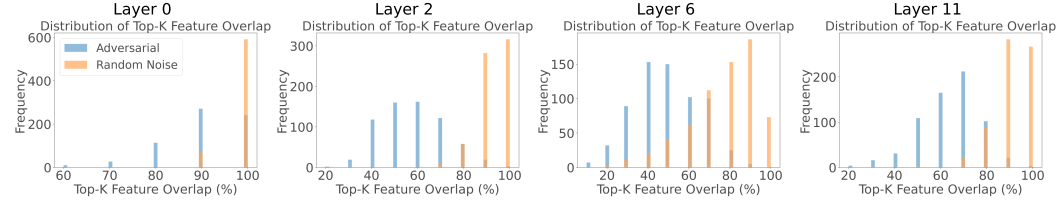


Figure 8: Feature overlap between original vs. attacked and original vs. noisy images for CLS token SAE features (threshold 0.1).

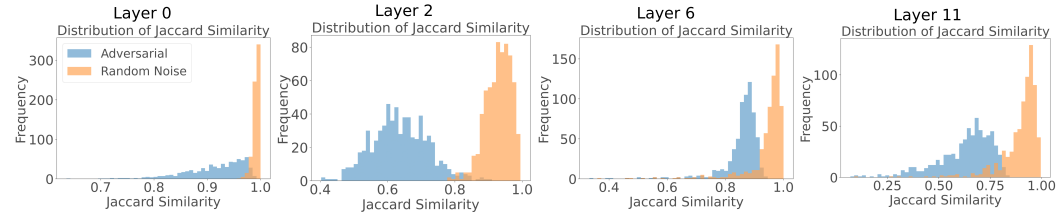


Figure 9: Comparison of Jaccard similarity for CLS token SAE features (threshold 0.1).

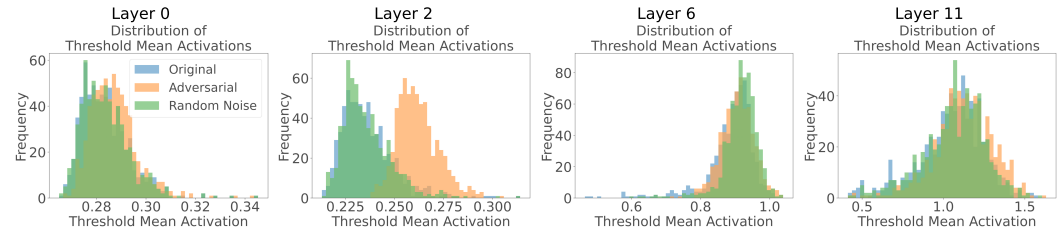


Figure 10: Histogram of mean thresholds for CLS token SAE features (threshold 0.1).

F.5 CLS TOKEN SAE FEATURES WITH LOW ACTIVATION THRESHOLD (0.001)

Here we examine SAE features from CLS tokens at a lower activation threshold of 0.001, so as to capture the broader population of features beyond only the most intensely activated ones. The subsequent figures show the histogram of activation counts (Fig. 11), the feature overlap (Fig. 12), the Jaccard similarity (Fig. 13), and the histogram of mean activations (Fig. 14) across layers for original, adversarial, and noisy inputs.

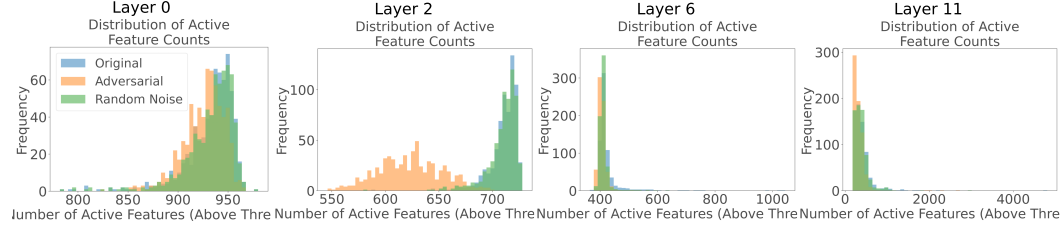


Figure 11: Histogram of activation counts for CLS token SAE features (threshold 0.001), with noise.

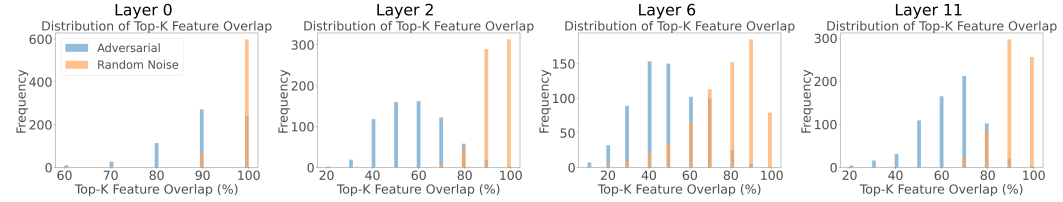


Figure 12: Feature overlap between original vs. attacked and original vs. noisy images for CLS token SAE features (threshold 0.001).

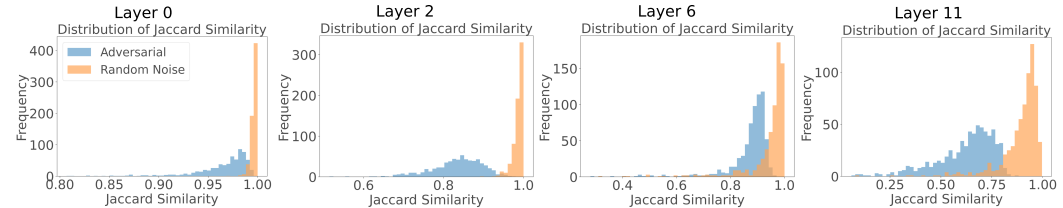


Figure 13: Comparison of Jaccard similarity for CLS token SAE features (threshold 0.001).

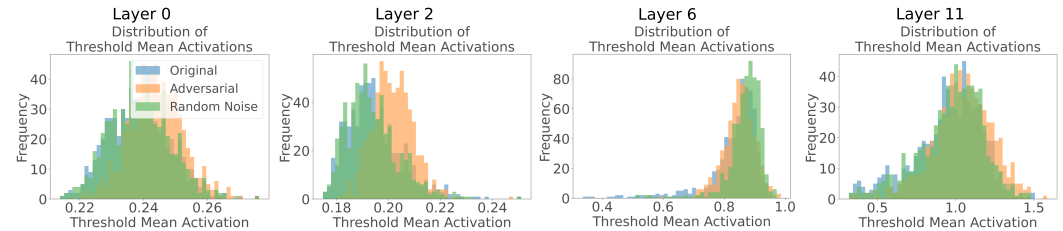


Figure 14: Histogram of mean activations for CLS token SAE features (threshold 0.001), with noise.

F.6 PATCH TOKEN SAE FEATURES WITH HIGH ACTIVATION THRESHOLD (0.1)

This subsection turns to SAE features derived from the mean of the patch tokens, again at a high activation threshold of 0.1. We present comparisons of the activation-count histograms (Fig. 15),

the distinct-feature-count histograms (Fig. 16), the feature overlap (Fig. 17), the Jaccard similarity (Fig. 18), and the mean-activation histograms (Fig. 19) for original, adversarial, and noisy inputs, thereby complementing the CLS-token analysis above.

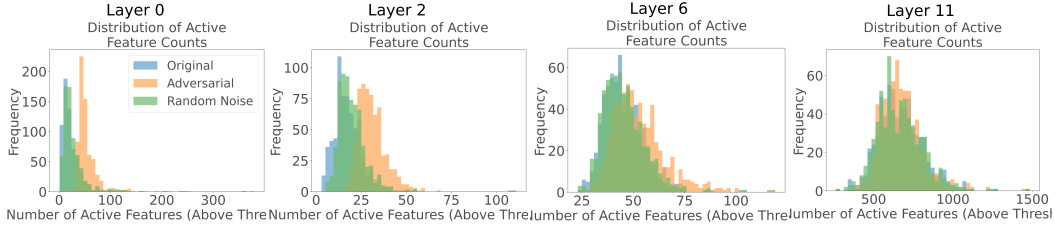


Figure 15: Histogram of activation counts for patch token SAE features (threshold 0.1), with noise.

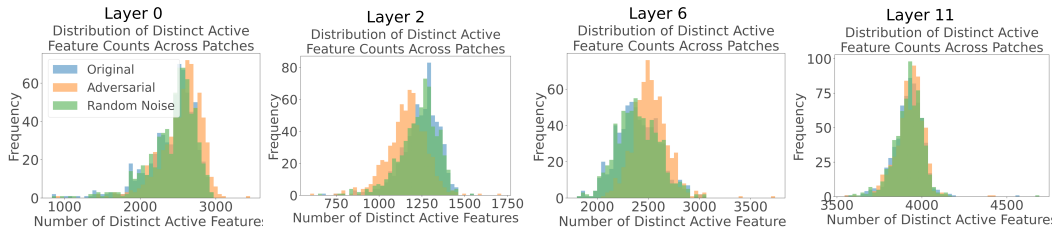


Figure 16: Histogram of distinct patch features for patch token SAE features (threshold 0.1), with noise.

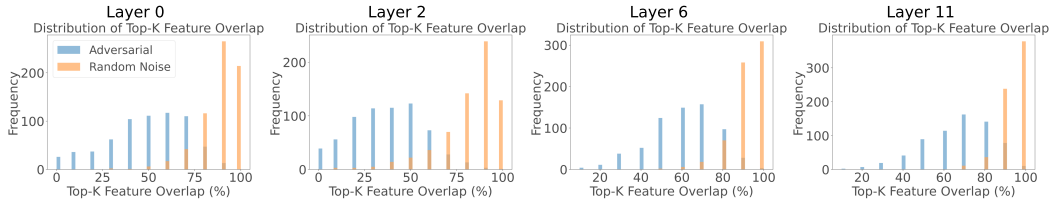


Figure 17: Feature overlap between original vs. attacked and original vs. noisy images for patch token SAE features (threshold 0.1).

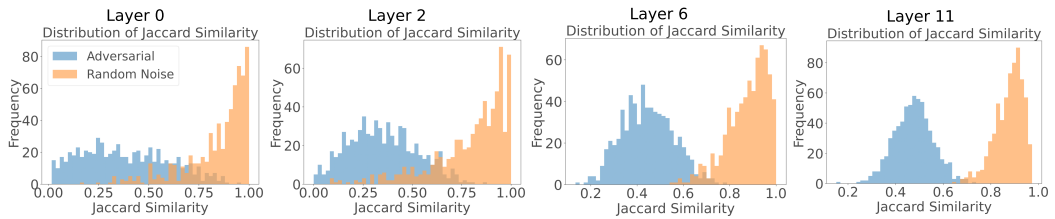


Figure 18: Comparison of Jaccard similarity for patch token SAE features (threshold 0.1).

F.7 PATCH TOKEN SAE FEATURES WITH LOW ACTIVATION THRESHOLD (0.001)

Finally, this subsection examines SAE features from the patch-token means at a low activation threshold of 0.001, thereby completing the four-way analysis over token type and threshold. The

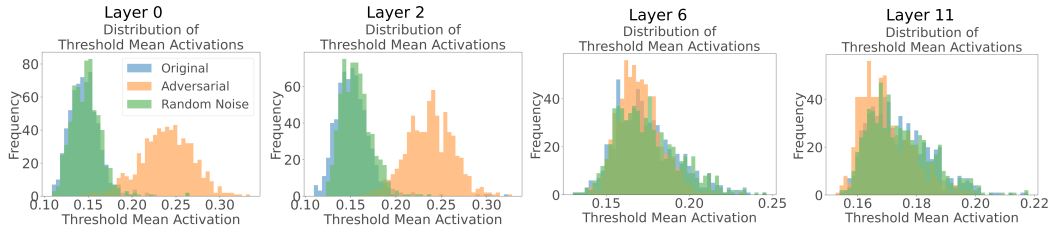


Figure 19: Histogram of mean thresholds for patch token SAE features (threshold 0.1), with noise.

figures show histograms of activation counts (Fig. 20), distinct feature counts (Fig. 21), feature overlap (Fig. 22), Jaccard similarity (Fig. 23), and mean activations (Fig. 24) for original, adversarial, and noisy inputs.

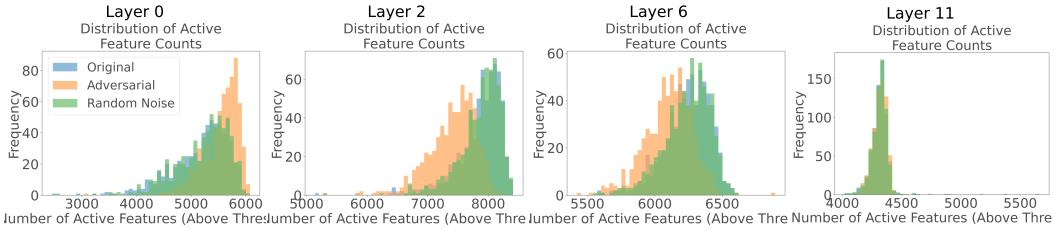


Figure 20: Histogram of activation counts for patch token SAE features (threshold 0.001), with noise.

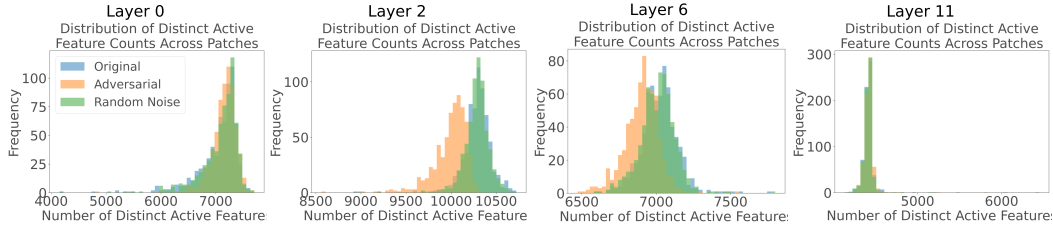


Figure 21: Histogram of distinct patch features for patch token SAE features (threshold 0.001), with noise.

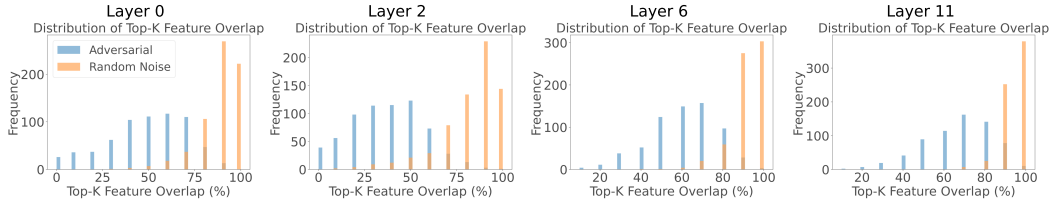


Figure 22: Feature overlap between original vs. attacked and original vs. noisy images for patch token SAE features (threshold 0.001).

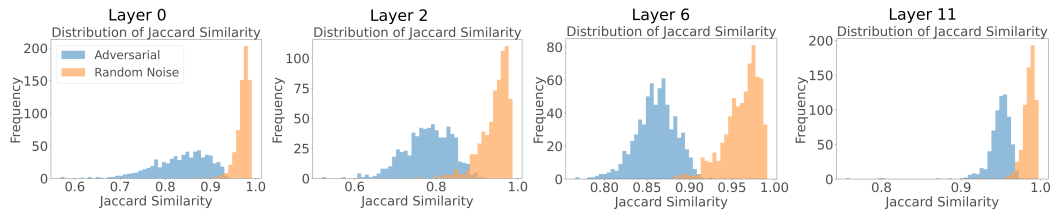


Figure 23: Comparison of Jaccard similarity for patch token SAE features (threshold 0.001).

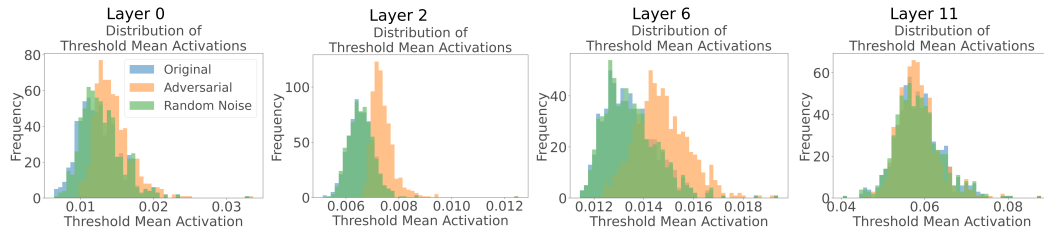


Figure 24: Histogram of mean activations for patch token SAE features (threshold 0.001), with noise.

G ACRONYMS

AEx adversarial example

CE cross entropy

LRH linear representation hypothesis

MLP Multilayer Perceptron

MSE mean squared error

NN neural network

PGD projected gradient descent

SAE sparse autoencoder

ViT vision transformer